

IDENTIFY THE STATES OF MATTER

All matter exists in one of three different physical forms: solid, liquid, or gas (Figure 7-5). When energy is added through increased temperature or pressure, or taken away through decreased temperature or pressure, changes in the states of matter occur. For example, water is a chemical compound that can exist in all three states of matter depending on its temperature. Consider how temperature changes create physical changes in water without changing its chemical properties:

- When water freezes, it turns to ice. As ice, water is in a solid state with definite shape and volume.
- When ice melts, it turns back into water. In its liquid state, water has volume, but no definite shape.
- When water boils, it turns to steam (vapor); when the steam cools, it turns back into water. As steam, water is a liquid that has evaporated into a gaslike state that does not have volume or shape.

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UNDERSTAND THE PHYSICAL AND CHEMICAL PROPERTIES OF MATTER

Every substance has unique properties that allow us to identify it. The two types of properties are *physical* and *chemical*.

Physical properties (FIZ-ih-kuhl PRAH-ur-teez) are characteristics that can be determined without a chemical reaction and that do not involve a chemical change in the substance. Physical properties include, but are not limited to, color, solubility, odor, density, weight, melting point, boiling point, and hardness.

Chemical properties (KEM-uh-kul PRAH-ur-teez) are characteristics that can only be determined by a chemical reaction and that cause a change in the identity of the substance. Rusting iron and burning wood are examples of a change in chemical properties. In both these examples, the chemical reaction of oxidation creates a chemical change in the substance: iron is changed to rust, and wood is changed to ash.

Physical and Chemical Changes

Matter can be changed in two different ways: Physical forces cause physical changes and chemical reactions cause chemical changes.

- A **physical change** (FIZ-ih-kuhl CHAYNJ) is a change in the form or physical properties of a substance without a chemical reaction or the creation of a new substance. Therefore, no chemical reactions are involved in physical change and no new chemicals are formed (Figure 7-6). An example of physical change is ice melting to water. Temporary haircolor is another example of physical change because it physically adds color to the surface of the hair, but does not create a chemical change in the hair's structure or color.

A **chemical change** (KEM-uh-kul CHAYNJ) is a change in the chemical composition or makeup of a substance, as in the iron-to-rust example. This change is caused by chemical reactions that create new chemical substances, usually by combining or subtracting certain elements, that will have

**DID YOU KNOW?**

Hydrogen is the most common element found in the known universe.

figure 7-5

Arrangement of particles in solid, liquid, and gaseous state.

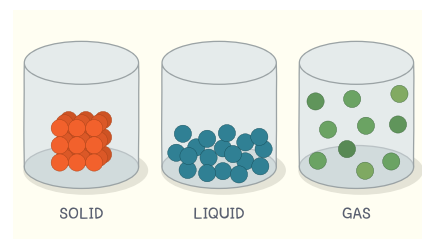


figure 7-6

Physical changes.

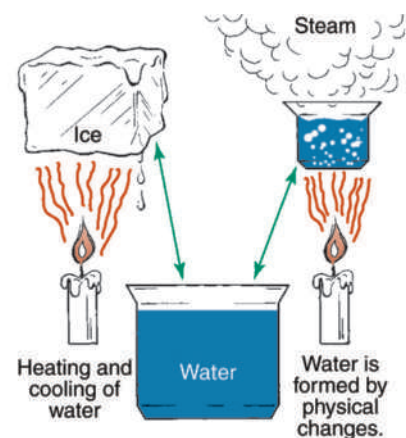
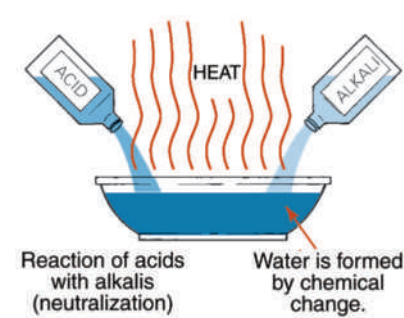


figure 7-7
Chemical changes.



different physical and chemical properties (Figure 7-7). An example of a chemical change is the oxidation of haircolor products. Oxidation refers to a chemical reaction that combines a substance with oxygen to produce an oxide. Permanent haircolor creates chemical changes in the hair when oxidation takes place to develop the dye in the color. In hair lightening processes, hydrogen peroxide oxidizes the melanin pigments in hair, leaving the hair a lighter color.

EXPLAIN OXIDATION-REDUCTION REACTIONS

Oxidation (ahk-sih-DAY-shun) can be defined as either the addition of oxygen or the loss of hydrogen. Oxidation is a chemical reaction that combines an element or compound with oxygen to produce an oxide. When oxygen is chemically combined with a substance, the substance is *oxidized*.

Oxidizing agents are substances that release oxygen. Hydrogen peroxide (H_2O_2) contains an extra oxygen atom that makes it an oxidizing agent. Different forms of hydrogen peroxide are commonly used in haircoloring, hair lightening, and chemical texture services.

Reduction (ree-DUK-shun) refers to either the loss of oxygen or the addition of hydrogen. When oxygen is chemically removed from a substance, the substance is *reduced*.

REDOX REACTIONS

Oxidation and reduction always occur simultaneously and are referred to as a **redox** (REE-dux) reaction. In a redox reaction, the oxidizing agent is always reduced and the reducing agent is always oxidized. Oxidation cannot occur without reduction. For example, when hydrogen peroxide (in the form of a developer) is mixed with an oxidation haircolor product (demi-permanent or permanent), the haircolor product gains oxygen and is oxidized. At the same time, the hydrogen peroxide loses oxygen and is reduced. In this example, the haircolor product is the reducing agent (Table 7-1).

Redox reactions also occur in another way that involves oxidation resulting from the loss of hydrogen and reduction that results from the addition of hydrogen. The permanent waving process is a good example of this type of redox reaction.

table 7-1
OXIDATION-REDUCTION REACTIONS

Product	Oxidation + Oxygen or – Hydrogen	Reduction – Oxygen or + Hydrogen
Oxidation Haircolor	Haircolor gains oxygen from H_2O_2	H_2O_2 loses oxygen
Permanent Wave Solution	Waving solution is oxidized	Adds hydrogen atoms to hair
Permanent Wave Neutralizer	Hair is oxidized by removing hydrogen	Neutralizer is reduced by losing oxygen
Exothermic Perm Activator	Adds oxygen to waving solution; creates heat	H_2O_2 loses oxygen

- Certain permanent waving solutions contain ammonium thioglycolate acid.
- The waving solution breaks the disulfide bonds in the hair through a reduction reaction that adds hydrogen atoms (H) to the hair. In this reaction, the hair is reduced and the permanent waving solution is oxidized.
- After processing and rinsing, the neutralizer is used to oxidize the hair by removing the hydrogen that was previously added with the waving solution. When the hair is oxidized, the neutralizer is reduced (Table 7-1).

Exothermic Reactions

When certain chemical reactions release energy in the form of heat, it is called an **exothermic reaction**. An example of this is the heat produced after mixing the activator and waving lotion in an exothermic permanent wave product. When the activator, which contains hydrogen peroxide, is added to the waving lotion, an oxidation reaction occurs that produces heat (Table 7-1). The mixing of these chemicals produces a more rapid form of oxidation than the slower oxidation that occurs with permanent wave neutralizers or oxidation haircolor products. In most cases, you can expect to feel a slight warming of the container after mixing the activator with the waving lotion.

Endothermic Reactions

An **endothermic reaction** is a chemical reaction that requires the absorption of energy or heat from an external source for the reaction to actually occur. Melting ice is an example of an endothermic reaction because if it was not absorbing heat from its surroundings, it would not be melting! Another example is a permanent waving lotion that requires the application of heat from a hood dryer to activate it for processing rather than processing at room temperature; these are called *endothermic waves*.

DEFINE PURE SUBSTANCES AND MIXTURES

All matter can be classified as either a pure substance or a physical mixture (blend) (Figure 7-8).

A **pure substance** (PYOOR SUB-stantz) is a chemical combination of matter that has a fixed chemical composition, definite proportions, and distinct properties. Elements and chemical compounds are pure substances. Aluminum foil is an example of a pure substance because it is composed only of atoms of the element, aluminum.

A **physical mixture** (FIZ-ih-kuhl mix-CHUR) is a combination of two or more substances united physically in any proportions without a fixed composition. Concrete is an example of a physical mixture because it is composed of water, sand, gravel, and dry cement. It is a mixture that has its own functions, yet does not lose the characteristics of the individual



REMINDER

When oxygen is combined with a substance, the substance is *oxidized*. When oxygen is removed from a substance, the substance is *reduced*.



DID YOU KNOW?

When performing chemical services, the term *oxidation* is used to define either the addition of oxygen or the loss of hydrogen. Conversely, the term *reduction* is used to define either the loss of oxygen or the addition of hydrogen. You will see these terms used again in Chapters 17 and 18.



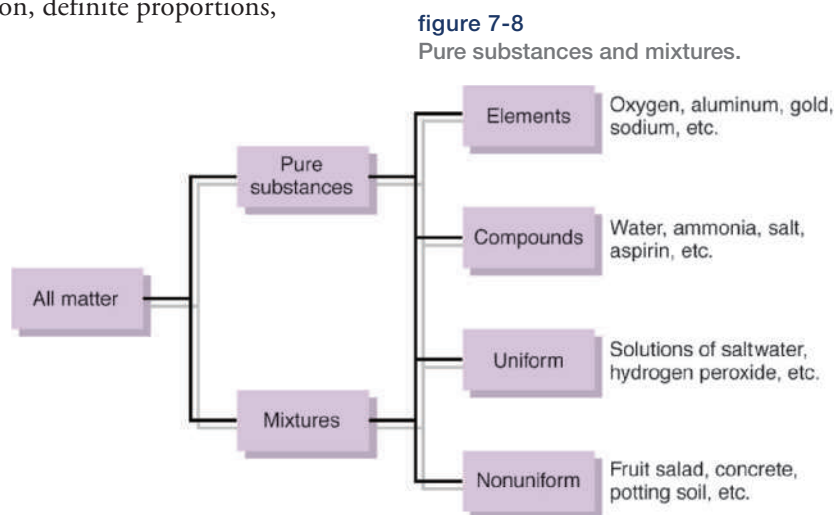
DID YOU KNOW?

When hydrogen peroxide is added to an oxidation haircolor, the temperature will increase due to the process of oxidation.



DID YOU KNOW?

Combustion is a form of *rapid oxidation* reaction that produces a high quantity of heat and light; lighting a match is an example of rapid oxidation.



ingredients. Powder is another good example of a physical mixture because it consists of a uniform mixture of insoluble inorganic, organic, and colloidal substances that have been properly blended.

IDENTIFY CHEMICAL COMPOUNDS

Chemical compounds (KEM-uh-kul KAHM-powndz) are combinations of two or more atoms of different elements united chemically with fixed chemical composition, definite proportions, and distinct properties. Chemical compounds are the result of a chemical reaction in which the elements that are united give up their own chemical identity and properties, for example, water is a chemical compound consisting of two atoms of hydrogen and one atom of oxygen. **Table 7-2** summarizes the differences between pure substances, physical mixtures, and chemical compounds.

As a barber, learning about chemical compounds will help your understanding of potential hydrogen and the pH scale. Chemical compounds are classified as follows:

1. **Oxides** are compounds of any element combined with oxygen. For example, one part carbon and two parts oxygen create carbon dioxide. One part carbon and one part oxygen create carbon monoxide.
2. **Acids** are compounds of hydrogen, a nonmetal such as nitrogen, and, sometimes, oxygen. For example, hydrogen, sulfur, and oxygen combine to form sulfuric acid. Acids turn blue litmus paper red, providing a quick way to test a compound.
3. **Bases**, also known as **alkalis** (AL-kuh-lyz), are compounds of hydrogen, a metal, and oxygen. For example, sodium, oxygen, and hydrogen form sodium hydroxide, which is used in the manufacture of soap. Bases will turn red litmus paper blue.
4. **Salts** are compounds that are formed by the reaction of acids and bases, with water also being produced by the reaction. Two common salts are sodium chloride (table salt), which contains sodium and chloride, and magnesium sulfate (Epsom salts), which contains magnesium, sulfur, hydrogen, and oxygen.

table 7-2
PURE SUBSTANCES, PHYSICAL MIXTURES, AND CHEMICAL COMPOUNDS

Pure Substances	Physical Mixtures	Chemical Compounds
Matter with a fixed chemical composition; elements, compounds, and elemental molecules	Two or more substances united physically with no fixed composition	Two or more different elements united chemically with a fixed chemical composition
Composition does not vary	Each substance retains its own identity and properties	Chemical reaction changes the chemical properties of the elements
Definite proportions	Mixed in any proportion	Definite proportions
Examples: oxygen, aluminum, gold	Examples: saltwater solution, pure air, concrete, powders	Examples: water (H ₂ O), salt (NaCl), ammonia

DEFINE SOLUTIONS, SUSPENSIONS, AND EMULSIONS

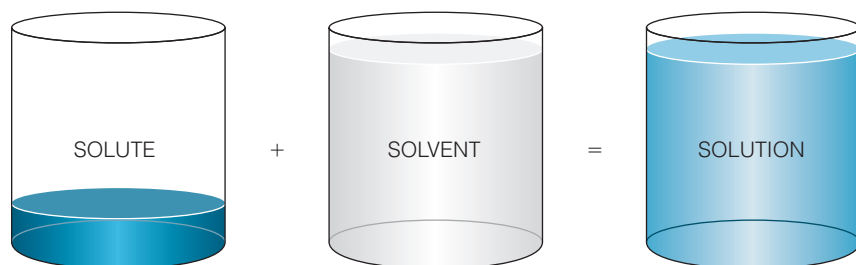
Solutions, suspensions, and emulsions are physical mixtures of two or more different substances. The differences between these mixtures are determined by the types of substances, the size of the particles, and the solubility of the substances. The *solubility* of a substance is defined as the amount of a substance that will dissolve in a given amount of another substance. For our purposes, we will define solubility to be the degree to which a solute (substance) dissolves in a solvent to make a solution.

All liquids are either miscible or immiscible. Miscible liquids, such as water and alcohol, are mutually soluble. This means they can be mixed together to form stable solutions. Conversely, immiscible liquids, such as water and oil, do not mix well together, are not capable of forming stable solutions, and will separate if left to sit for any time.

Solutions

A **solution** is a stable, uniform mixture of two or more mixable substances that is made by dissolving a solid, liquid, or gaseous substance in another substance. A **solute** is any substance that is dissolved into a solvent to form a solution. A **solvent** is any substance, usually a liquid, that dissolves the solute to form a solution. For example, dissolving salt (solute) into water (solvent) creates a saltwater solution. The components of a solution are illustrated in **Figure 7-9**. When a gas or a solid is dissolved in a liquid, the gas or solid is the solute and the liquid is the solvent.

figure 7-9
Components of a solution.



Suspensions

Suspensions (sus-PEN-shunz) are unstable uniform mixtures of two or more substances. The particles in suspensions can be seen with the naked eye because they are larger than the particles in solutions.

- Suspensions are not usually transparent, may be colored, and tend to separate over time.
- Many of the products used by barbers, such as hair tonics, are suspensions and should be shaken or mixed well before use. Calamine lotion and paint are also examples of suspensions.

Emulsions

An **emulsion** is a physical mixture of two immiscible liquids held together by an emulsifying agent. An emulsion is a suspension of one liquid dispersed in another. Although emulsions tend to separate over

table 7-3

SOLUTIONS, SUSPENSIONS, AND EMULSIONS

Solutions	Suspensions	Emulsions
Miscible	Slightly miscible	Immiscible
No surfactant	No surfactant	Surfactant
Small particles	Larger particles	Largest particles
Usually clear	Usually cloudy	Usually a solid color
Stable mixture	Unstable mixture	Limited stability
Witch hazel	Calamine lotion	Shampoos and conditioners

time, with proper formulation and storage an emulsion may remain stable for at least 3 years.

- Mayonnaise is an oil-in-water emulsion of two immiscible liquids. The egg yolk in mayonnaise emulsifies the oil droplets and disperses them uniformly in the water. Mayonnaise should not separate upon standing. Review [Table 7-3](#) for a summary of the differences between solutions, suspensions, and emulsions.
- Shampoos and conditioners are examples of emulsions used in barbering services.
- Cosmetic emulsions should be used within one year of purchase.

Surfactants

A **surfactant** (sur-FAK-tant) (or base detergent) is a substance that allows oil and water to mix or emulsify by reducing surface tension. The term *surfactant* is a contraction for *surface active agent*.

A surfactant molecule has two distinct parts: the head of the molecule is hydrophilic (water-loving) and the tail is lipophilic (oil-loving) ([Figure 7-10](#)). Since “like dissolves like,” the hydrophilic head dissolves in water and the lipophilic tail dissolves in oil. This configuration allows the surfactant molecule to mix and dissolve in both water and oil to join together to form an emulsion.

- In an oil-in-water (O/W) emulsion, droplets of oil are dispersed in water, where they are surrounded by surfactants. The lipophilic surfactant tails point inward with the hydrophilic heads pointing outward, which keeps the oil dispersed in the water. Most of the emulsions used in the barbershop are O/W emulsions and do not feel as greasy as water-in-oil preparations.

figure 7-10
Surfactant molecules.



- In a water-in-oil (W/O) emulsion, droplets of water are dispersed in oil, where they are surrounded by surfactants. The surfactant tails point outward and the heads point inward, which keeps the water dispersed in the oil. Cold cream is an example of a W/O emulsion.

Discuss the Properties of Water and pH

After reading this section, you will be able to:

- LO 6** Define *pH* and describe the pH scale.
- LO 7** Explain how product pH levels affect the hair and skin.

Water (H₂O) is the most abundant and important of all chemicals, composing about 71 percent of the earth's surface and an average of about 65 percent of the human body. Water is known as the *universal solvent*, because it dissolves more substances than any other liquid. Distilled or demineralized water is used as a nonconductor of electricity, while water containing certain mineral substances is an excellent conductor of electricity.

Water is purified through boiling, filtration, or distillation.

- Boiling water at 212 degrees Fahrenheit destroys most microbes and renders it suitable for drinking.
- During filtration, water passes through a porous substance, such as filter paper or charcoal, to remove organic material.
- Distillation is the process whereby water is heated to a vapor in a closed vessel. The vapors are captured and passed off through a tube into another vessel, where they are cooled and condensed to a liquid. This type of purified water is used in the manufacturing of cosmetics.

Soft water is rainwater or chemically treated water that has low levels of mineral substances such as calcium and magnesium salts. This type of water not only allows soaps and shampoos to lather freely but can also leave the hair and skin feeling too slick even after thorough rinsing.

Conversely, hard water contains mineral substances, such as calcium and magnesium salts, that can interfere with the lathering and foaming process. Hard water may be softened by distillation, water-softening units, or by adding sodium carbonate (washing soda); however, today's hair products and modern filtration systems have helped eliminate some of the problems associated with hard water in the past. If you are in doubt about the quality of the water in the barbershop, use a test kit designed to measure both the water hardness and pH level.

LEARN ABOUT WATER AND pH (POTENTIAL HYDROGEN)

A basic understanding of ions is important to an understanding of pH.

An **ion** (EYE-ahn) is an atom or molecule that carries an electrical charge.

Ionization (eye-ahn-ih-ZAY-shun) is the separation of a substance into ions



DID YOU KNOW?

Seventy percent of the earth's surface is covered in water; the human body is 65 percent water.

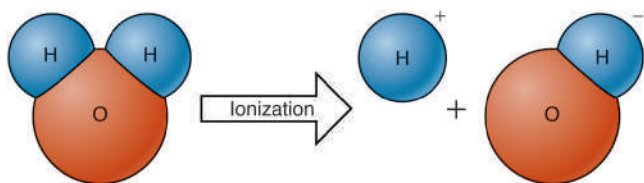


DID YOU KNOW?

You can test water to determine whether or not it is soft by using a soap solution. Use the following steps to test the water softness in your area.

1. Dissolve $\frac{3}{4}$ of an ounce of pure, powdered castile soap in a pint of distilled water.
2. Use a second pint bottle and fill it halfway with tap water.
3. Add about 7 drops of the soap solution.
4. Shake the bottle vigorously. If lather forms at once and persists, the water is very soft.
5. If lather does not appear at once, add another 7 drops or ounces of soap solution. Shake again. If an additional 7 drops or ounces of the soap solution is needed to produce a good lather, the water is hard and should be softened for hair services in the barbershop.

figure 7-11
ionization of water.



DID YOU KNOW?
The optimum pH level of the body fluids is 7.4. Minerals in food and water help balance the acidity and alkalinity of these fluids in the human body.

that have opposite electrical charges. A negatively charged ion is called an *anion*, and a positively charged ion is called a *cation*.

In pure water, some of the water molecules ionize naturally into hydrogen ions and hydroxide ions. The pH scale measures those ions. The hydrogen ion is acidic and the hydroxide ion is alkaline.

- The ionization of water is what makes pH possible because only aqueous (water) solutions have pH. Nonaqueous solutions, such as alcohol or oil, do not have pH.
- In pure water, every water molecule that ionizes produces one hydrogen ion and one hydroxide ion (**Figure 7-11**). Pure water contains the same number of hydrogen ions as hydroxide ions, which makes it neutral, as denoted by the number 7 on the pH scale. This means that pure water is 50 percent acidic and 50 percent alkaline.
- The letters **pH** (P-H) denote **potential hydrogen**, which is the relative degree of acidity or alkalinity of a substance. The **pH scale** (P-H SKAYL) measures the concentration of hydrogen ions in acidic and alkaline *water-based* solutions. Notice that pH is written with a small *p*, which represents quantity, and a capital *H*, which represents the hydrogen ion.

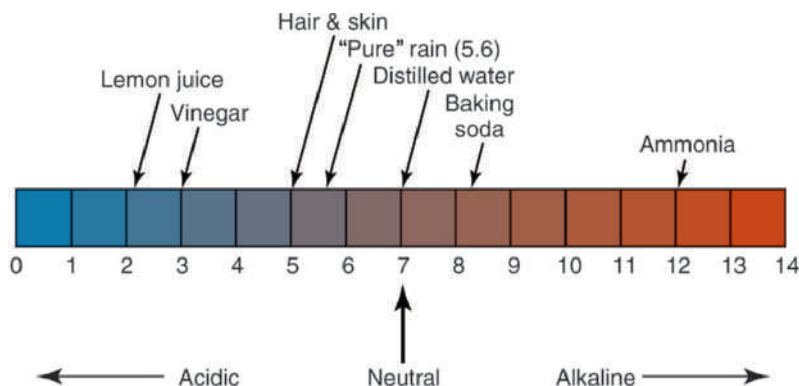
The pH Scale

The pH values are arranged on a scale ranging from 0 to 14. A pH below 7 indicates an acidic solution, a pH of 7 is a neutral solution, and a pH above 7 indicates an alkaline solution (**Figure 7-12**).

The pH scale is a logarithmic scale, which means that a change of one whole number represents a 10-fold change in pH. For example, a pH of 8 is 10 times more alkaline than a pH of 7. A change of two whole numbers indicates a change of 10 times 10, or a 100-fold change. Therefore, a pH of 9 is 100 times more alkaline than a pH of 7.

The pH range of hair and skin is 4.5 to 5.5, with an average of 5, on the pH scale. That means that pure water is 100 times more alkaline than hair and skin, even though it has a neutral pH. In fact, pure water can cause the hair to swell up to 20 percent.

figure 7-12
The pH scale.



Acids and Alkalis

All acids owe their chemical reactivity to the hydrogen ion (H^+).

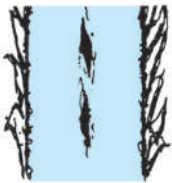
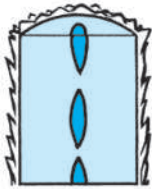
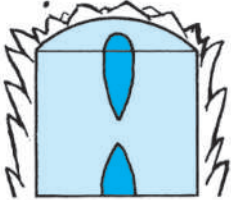
- Acids have a pH below 7.0, taste sour, and turn litmus paper from blue to red.
- Acidic solutions tend to contract and harden the hair (Figure 7-13). As toners or skin fresheners, they also tighten the skin.

All alkalis owe their chemical reactivity to the hydroxide (OH^-) ion. The terms *alkalis* and *bases* are interchangeable.

- Alkalis have a pH above 7.0, taste bitter, feel slippery on the skin, and turn litmus paper from red to blue.
- Alkalis soften and swell the hair (Figure 7-13). Sodium hydroxide (lye) is a very strong alkali used in chemical drain cleaners and chemical hair relaxers.

Figure 7-13

The effect of pH on hair.

Solution	Effect on Hair		Important Features
Very Strong Acid (pH 0.0–1.0)		Dissolves hair completely.	Must not be applied to hair or scalp.
Strong to Mild Acid (pH 1.0–4.5)		Hair shrinks and hardens. Body is increased. Cuticle imbrications close up. Porosity is reduced. Sheen of hair is improved. Soap residues are removed. Neutralizes traces of alkalis.	Acid or cream rinses restore body to bleached, porous hair. Conditioners and fillers overcome the excess porosity of damaged hair. Special shampoos reduce tangling and matting of hair and prevent color loss. Hair creams increase sheen. Color rinses provide temporary effect. Neutralizers remove residual waving lotion.
Neutral (pH 4.5–5.5)		Hair is normal diameter. Texture and luster standard.	Neutral solutions are designed to prevent excess swelling of normal and damaged hair. Mild shampoos for normal cleaning and manageability of hair.
Mild Alkali (pH 5.5–10.0)		Hair swells. Porosity increases as imbrications open. Hair has a dry, drab appearance.	Tints and bleaches penetrate easier and chemical action increases. Cold wave solutions for resistant hair. Soap shampoos to overcome acidity of tap water. Activators for hydrogen peroxide.
Stronger Alkali (pH 10.0–14.0)		Dissolves hair completely.	Must not be applied to hair or scalp unless used as relaxers or depilatories.

Acid–Alkali Neutralization Reactions

When acids and alkalis are mixed together in equal proportions, they neutralize each other to form water and a salt. For example, because hydrochloric acid is a strong acid and sodium hydroxide is a strong alkali, they will neutralize each other when mixed in equal amounts and form a solution of pure water and table salt (refer to [Figure 7-7](#)). Acid-balanced shampoos and normalizing lotions associated with hydroxide hair relaxers work to create a similar acid–alkali neutralization reaction.

Identify Cosmetic Preparations Used in Barbering

After reading this section, you will be able to:

- LO⁸ Name nine types of shampoos.
- LO⁹ List four classifications of conditioners.
- LO¹⁰ Recognize other cosmetic preparations used in barbering services.

LEARN ABOUT SHAMPOOS AND CONDITIONERS

Most shampoo products are emulsions, and there are many different types on the market. As a professional barber, you will need to become skilled at selecting products that best serve the condition of the client's hair and scalp. Make it a standard operating procedure to read the manufacturer's labels so that an informed choice can be made. In addition, display and use the barbershop's retail products at the workstation or shampoo sink back bar. Doing so not only promotes product sales but also demonstrates to clients that you endorse the product and its effectiveness inside and outside the barbershop.

To be effective, a **shampoo** (sham-POO) must remove all dirt, oil, perspiration, and skin debris, without adversely affecting either the scalp or the hair. The hair collects dust particles, natural oils from the sebaceous glands, perspiration, and dead skin cells that can accumulate on the scalp. This accumulation creates a breeding ground for disease-producing bacteria, which can lead to scalp disorders. The hair and scalp should be thoroughly shampooed as frequently as is necessary to keep it clean, healthy, and free from bacteria.

An effective shampoo product should:

- cleanse the hair of oils, debris, and dirt.
- work efficiently in hard as well as soft water.
- not irritate the eyes or skin.
- leave the hair and scalp in their natural condition.

Shampoo Chemistry

The acidity or alkalinity of a shampoo is important because it influences how the product will affect various layers of the hair and skin. Acidic solutions (below pH 7.0) shrink, constrict, and harden the cuticle scales of the

hair shaft. An alkaline solution (above pH 7.0) softens, swells, and expands the cuticle scales. Remember, the lower the pH of a solution, the greater the degree of acidity; the higher the pH of a solution, the greater the degree of alkalinity. (See **Figure 7-12**.)

Shampoo emulsions usually range between 4.5 and 7.5 on the pH scale. Since the normal pH range for hair and skin is 4.5 to 5.5, mild or more acidic shampoos are found closer to this range. Conversely, stronger or more alkaline shampoos are found beyond 6.0 on the pH scale.

Shampoo Molecules

Shampoos consist of two main ingredients: water and surfactants. Next to water, surfactants are the second most common ingredients found in shampoos and act as cleansing, emulsifying, or foaming agents. Shampoo molecules are created by combining water and at least one surfactant. In a shampoo molecule, the tail attracts dirt, grease, debris, and oil, but repels water. The head of the shampoo molecule attracts water, but repels dirt. Working together, both parts of the molecule effectively cleanse the hair (**Figures 7-14 to 7-17**).

The base surfactant or combination of surfactants determines the classification of a shampoo. These classifications include *anionic*, *cationic*,

figure 7-14

The tail of the shampoo molecule is attracted to oil and dirt.

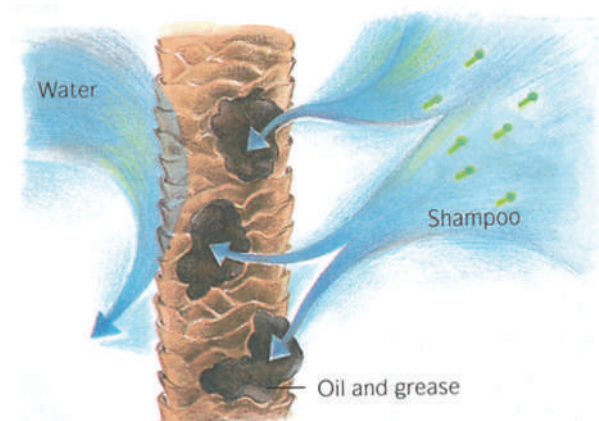


figure 7-15

Shampoo causes oil to run up into small globules.

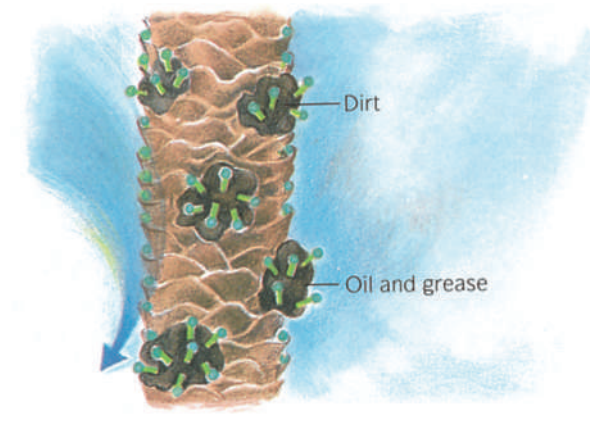


figure 7-16

During rinsing, the heads of the shampoo molecules attach to water molecules and cause debris to roll off.

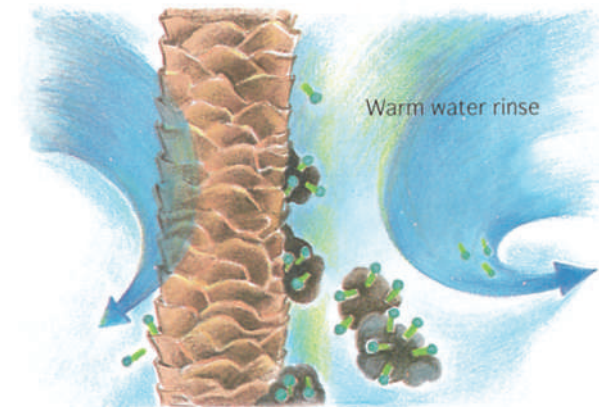


figure 7-17

Thorough rinsing washes away debris and excess shampoo.

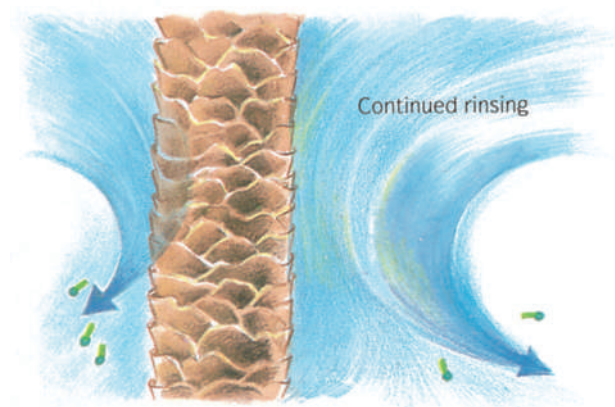


table 7-4

CLASSIFICATIONS OF SHAMPOOS

Base Surfactant(s)	Common Ingredients	Actions	Features
Anionics	Sodium lauryl sulfate	Relatively harsh cleanser; produces rich foam	Rinses easily
	Sodium laureth sulfate	Less alkaline than lauryl	Less drying to hair shaft
Cationics	Quaternary ammonium compounds	Antibacterial action	Used in dandruff shampoos
Nonionics	Cocamide (DEA, MEA)	Mild cleansing action	Low incidences of irritation
Amphoterics	Amphoteric I-20	Behaves as anionic or cationic depending on pH; clings to hair for manageability	Nonstinging to eyes; used in baby shampoos

nonionic, and *amphoteric* surfactants. Most manufacturers use detergents from more than one classification. It is customary to use a secondary surfac-
tant to complement, or offset, the negative qualities of the base surfactant.
For example, an amphoteric that is nonirritating to the eyes can be added to
a harsh anionic to create a product that is more comfortable to use.

Additional ingredients in the forms of moisturizers, preservatives, foam
enhancers, perfumes, and others are added to create a variety of shampoo
formulations. Along with water and base surfactants, these ingredients are
listed on product labels in descending order according to the percentage of
each ingredient in the shampoo. Refer to **Table 7-4** to review the properties
associated with different shampoo classifications.

IDENTIFY TYPES OF SHAMPOOS

Shampoo products account for the highest dollar expenditure by consumers
in hair care products. Shampoo products are available in liquid, liquid-dry,
and dry (powder) formulations to meet the needs of all types of hair and
scalp conditions. There are shampoos for dry, oily, normal, fine, coarse, limp,
or chemically treated hair. There are shampoos that add a slight amount of
color to hair and those that cleanse the hair of mineral deposits and product
buildup. Many manufacturers have become more “hair condition conscious”
so that even medicated shampoos are less drying to the hair and scalp.

A visit to your local barber supply will introduce you to a wide variety
of products for use in the barbershop. Be sure to read the label for the
product’s recommended use and refer to **Table 7-4** to become familiar with
base surfactants.

- Most of the shampoo products you use will be in a cream or clear liquid
form. Generally speaking, cream shampoos contain more emollients
than clear shampoos.
- **Liquid-dry shampoos** are used for cleansing the scalp and hair when the cli-
ent is prevented from having a regular shampoo. Typically, these shampoos
have an astringent quality that allows the product to evaporate quickly.

- **Dry or powder shampoos** can be used when the client's health or condition will not permit a wet shampoo. The powder is sprinkled onto the hair, where it picks up dirt and oils as it is brushed through the hair. Always follow the manufacturer's directions and never give a dry shampoo prior to a chemical service.

Most of today's shampoos can be classified as belonging to one of the following shampoo types.

pH-Balanced Shampoos

pH-balanced shampoos, also known as *acid-balanced shampoos*, are balanced to the pH level of hair and skin and usually contain citric, lactic, or phosphoric acid. These shampoos are mild formulations designed to prevent the stripping of haircolor from the hair. They have a low alkaline content, which makes them a good choice for normal, chemically treated, and fragile hair. Follow the manufacturer's directions, but most pH-balanced shampoos can be used daily.

Balancing Shampoos

Balancing shampoos are designed for oily hair and scalps. These shampoos wash away excess oiliness while keeping the hair from drying out.

Clarifying Shampoos

Clarifying shampoos contain an active chelating agent that binds to metals (such as iron and copper) and removes them from the hair. They also provide thorough cleansing and cut through product buildup that can coat and flatten hair. For these reasons, clarifying shampoos are recommended for preparing the hair for chemical services or to remove medication or hard-water minerals from the hair.

Color-Enhancing Shampoos

Color-enhancing shampoos are created by combining the surfactant with basic colors. They are attracted to porous hair and provide only slight color changes, which are removed with plain shampooing. Color shampoos are used to brighten the hair, add slight color, and eliminate unwanted color tones.

Conditioning Shampoos

Conditioning shampoos (KAHN-dish-shun-ing sham-POOZ), also known as *moisturizing shampoo*, are mild cream shampoos that contain moisturizing agents (*humectants*) designed either to "lock in" the moisturizing properties of the product or to draw moisture into the hair. They are formulated to make the hair smooth and shiny and to avoid damaging chemically treated hair. Protein and biotin are conditioning agents that help restore moisture and elasticity, strengthen the hair shaft, and add volume. Like pH-balanced shampoos, moisturizing shampoos are non-stripping shampoos and do not remove artificial color from the hair.

Medicated Shampoos

Medicated shampoos contain antifungal agents such as pyrithione zinc, selenium sulfide, or ketoconazole, which help control dandruff by suppressing the growth of *malassezia*, the fungus that causes dandruff. These shampoo products are available for all hair types and are gentle enough for daily use.

Therapeutic medicated shampoos (thayr-uh-PYOOT-ik MED-ih-kayt-ud sham-POOZ) contain special chemicals or drugs that are very effective in reducing excessive dandruff. These are prescribed shampoos that must include instructions from the client's physician for their application and use.

Neutralizing Shampoos

A **neutralizing shampoo** helps to rebalance and restore the pH level of the hair by neutralizing any alkali or unwanted residues that remain in the hair after chemically relaxing the hair. Some manufacturers call their neutralizing shampoo, *balancing* shampoo; be sure to read labels carefully to avoid confusing this type of balancing shampoo with a balancing shampoo intended for oily hair and scalp.

Sulfate-Free Shampoos

Sulfates, such as sodium laureth sulfate, are surfactants used in shampoos and other products for their thickening and lathering capabilities. A **sulfate-free shampoo** is formulated with little to no sulfates and is generally a good choice for color-treated and chemically treated hair because it does not strip natural oils from the hair.

Shampoos for Hair Replacements and Wigs

Wig and hair replacement shampoos are specially prepared products for cleaning hair systems. You should always use the manufacturer's recommended cleaning product because some hair systems require the use of a cleaning solvent rather than the mild shampoo that may be appropriate for other hair replacement products. Always follow the manufacturer's instructions carefully.

IDENTIFY TYPES OF CONDITIONERS

Conditioners are special chemical agents that are applied to hair to deposit protein or moisture to help restore the hair's strength and body or to protect against possible damage. Conditioners typically range from 3.0 to 5.5 on the pH scale. They are a temporary remedy for hair that feels dry or appears damaged. As such, conditioners cannot actually repair damaged hair nor can they increase or improve the quality of new hair growth.

Excessive use of conditioners, or using the wrong type of conditioner, can lead to a heavy or oily buildup of product on the hair. For this reason, you need to be able to select the right product for your client, for example, an instant conditioner for detangling or a repair conditioner for damaged hair.

There are three basic types of conditioners: instant (or rinse-out), treatment or repair, and leave-in. As a barber, you must decide the type to use based on the texture and condition of your client's hair and the desired results.

- **Instant conditioners** (IN-stant KAHN-dish-uhn-uhrz), also known as *rinse-out conditioners*, are applied to freshly shampooed hair and rinsed out after being worked through the hair. These conditioners usually have a lower pH than the hair, which helps close the cuticle scales. The typical instant conditioner does not penetrate into the hair shaft but may add oils, moisture, and sometimes protein to the cuticle scales. Detangling rinses, basic conditioners, and finishing rinses are examples of instant conditioners.

- **Treatment or repair conditioners** are deep, penetrating formulations that help restore moisture and protein in the hair. These conditioners sometimes require longer processing time or the application of heat before being rinsed out.
- **Leave-in conditioners** (LEE-v-in KAHN-dish-uhn-uhz), such as thermal protectors or blowdrying sprays, are applied to hair and not rinsed out. Some of these conditioners are designed for use with thermal tools, while others may be included with permanent wave products to help equalize the porosity of the hair shaft.
- **Medicated conditioners**, or rinses, are formulated to help control minor dandruff and scalp conditions. A dandruff rinse is a commercial product that usually works in conjunction with an antidandruff shampoo to help control dandruff. Always follow the manufacturer's directions.

Conditioner Ingredients

Most conditioners contain a large amount of fatty materials and a small amount of surfactant. In fact, without surfactants, conditioners could not be rinsed from the hair. In addition, conditioners contain the following types of ingredients:

- Protein and protein derivatives consist of hydrolyzed protein that can pass through the cuticle and penetrate into the cortex of the hair. Concentrated protein conditioners are used to increase the tensile strength of the hair and to temporarily close split ends. They also improve texture, equalize porosity, and help increase elasticity in the hair. Deep-conditioning treatments, also known as *hair masks* or *conditioning packs*, are chemical mixtures of concentrated protein in the heavy cream base of a moisturizer. Be sure to rinse the hair thoroughly prior to cutting, setting, or drying, as excess conditioner may coat or weigh down the hair.
- Fatty materials are the basic ingredients of many conditioners and include fatty alcohols, silicones, humectants, and moisturizers.
 - *Fatty alcohols* consist of nonvolatile oils, fats, or waxes that contain an alcohol group. Some examples are cetyl alcohol, glycerol, and stearyl alcohol.
 - *Humectants* are chemical compounds that attract and retain moisture from the atmosphere. For example, moisturizing conditioners contain humectants that help seal moisture inside the hair by coating the cuticle.
 - *Moisturizers* frequently refer to oily substances that coat the hair or skin and prevent water loss by evaporation. Lanolin, mineral oils, and cholesterol are examples of water-trapping additives found in many conditioners.
 - *Silicones* are oils that contain a repeating silicon-oxygen chain that is superior to plain oils because they are less greasy and form a breathable film over the hair or skin.

It is the barber's responsibility to be knowledgeable about the products used in the shop or salon. Basic product knowledge can be easily obtained from product labels, distributors, trade show demonstrations, and manufacturer representatives. See **Table 7-5** for shampoo and conditioning products suitable for different hair types.

table 7-5

MATCHING PRODUCTS TO HAIR TYPES

Hair type	Fine	Medium	Coarse
Straight	Volumizing shampoo Detangler, if necessary Protein treatments Fine-hair shampoo	pH-balanced shampoo Finishing rinse Protein treatments	Moisturizing shampoo Leave-in conditioner Moisturizing treatments
Wavy, Curly, Extremely Curly	Light leave-in conditioner Protein treatments Spray-on thermal protectors	pH-balanced shampoo Leave-in conditioner Moisturizing treatments	Leave-in conditioner Protein and moisturizing treatments
Dry and Damaged (perms, color- treated, relaxers, blowdrying, sun, hot irons)	Gentle cleansing shampoo Light leave-in conditioner Protein and moisturizing repair treatments Spray-on thermal protection	Shampoo for chemically treated hair Moisturizing conditioner Protein and moisturizing repair treatments	Deep-moisturizing shampoo for damaged hair Leave-in conditioner Deep-conditioning treatments and hair masks

Scalp Conditioners

Scalp conditioners are topical preparations designed to maintain or improve the condition of the scalp. These products are available in a variety of formulations for different purposes; however, none will make the hair “grow.”

- Cream-based products with moisturizers and emollients are usually used to soften and improve the health of the scalp.
- Medicated scalp lotions or ointments are products that promote healing of the scalp and may be prescribed by a physician.
- Astringent scalp tonics help remove oil accumulation on the scalp and can be used after a scalp treatment.

BECOME FAMILIAR WITH STYLING AIDS

Styling aids are available in a variety of formulations that include hair sprays, hair creams, tonics, waxes, gels, mousses, molding pastes and clays, to name but a few. The main thing to remember about using these products is to become familiar with their properties so that you can select the right product for the job. Some products will provide too much control and others not enough. For example, while a hair spray may be effective for holding a finished style in place, your client may prefer a light cream product that allows his or her hair to move more freely. With practice and experience you will identify the products that best meet your clients’ needs.

Hair Tonics

Hair tonics are liquid or cream preparations, generally considered to be grooming or styling aids, that have been a part of the barbershop environment for decades. Depending on the formula, these products are used to stimulate the surface circulation of the scalp; remove loose dandruff; or impart manageability, shine, and control to the hair. Hair tonics are leave-in products that are available in oil-based, water-based, and

alcohol-based formulations. Be sure to read the label carefully to select the right product for the intended purpose.

REVIEW OTHER COSMETIC PREPARATIONS

As a barber, you will use a variety of cosmetic preparations as you perform services on clients. The following section provides a brief description of some of the other products that you should become familiar with in preparation for working in a barbershop.

- *Astringents* may have an alcohol content of up to 35 percent. Astringents cause contraction of tissues and may be used to remove oil accumulation on the skin or to close the pores after a facial or shave. Due to the high alcohol content of astringent lotions, some skin types may be sensitive and react with a slight swelling or redness of the skin. Discontinue use if either of these symptoms appears. Skin tonics or toners and fresheners are milder forms of astringents.
- *Skin tonics or toners* usually have an alcohol content of 4 to 15 percent, which positions them between fresheners and astringents in terms of alcohol content. Most toners are designed for use on normal and combination skin types.
- *Fresheners*, also known as *skin freshening lotions*, have the lowest alcohol content (0 to 4 percent) in the group of astringents. They are designed for dry, mature, and sensitive skin types. The formulation of a freshener typically includes some or all of the following: witch hazel, alcohol and camphorated alcohol, citric acid, boric acid, lactic acid, phosphoric acid, aluminum salts, menthol, chamomile, and floral scents.
- *Cleansing creams* are used during facials and shaves in the barbershop. The action of a cleansing cream is caused in part by the oil content of the cream, which has the ability to dissolve other greasy substances. Older formulas, such as cold cream, contain relatively few ingredients that may include vegetable or mineral oil, beeswax, water, preservatives, and emulsifiers. The newer cleansing formulations are more skin condition-specific and may contain additional degreasers such as lemon juice, synthetic surfactants, emollients, or humectants.
- *Cleansing lotions* serve the same purposes as cleansing creams but are of a lighter consistency and are usually water-based emulsions. They are available in dry-, normal-, and oily-skin formulations. Some ingredients common to cleansing lotions are cetyl alcohol, cetyl palmitate, and sorbitol; perfumes and colorings are added to enhance lotions' marketing value.
- *Depilatories* are preparations used for the temporary removal of superfluous hair by dissolving it at the skin line. Depilatories contain detergents to strip the sebum from the hair and adhesives to hold the chemicals to the hair shaft for the 5 to 10 minutes necessary to remove the hair. During the short processing time, swelling accelerating agents such as urea or melamine expand the hair, helping break the hair bonds. Finally, chemicals such as sodium hydroxide, potassium hydroxide, thioglycolic acid, or calcium thioglycolate destroy the disulfide bonds. These chemicals turn the hair into a soft, jellylike mass of hydrolyzed protein that can be scraped from the skin. Although depilatories are

not commonly used in barbershops, knowing about them is necessary because your customers may use them at home.

- *Epilators* remove hair by pulling it out of the follicle. Two types of wax are currently used for professional epilation: cold and hot. Both products are made primarily of resins and beeswax. Beeswax has a relatively high incidence of allergic reaction; therefore, it is advisable to do a small patch test of the product to be used. Recently, an electrical apparatus made for the home hair-removal market has become available.
- *Hair spray* is used to hold the finished style. Many new formulations for hair spray contain a variety of polymers, such as acrylic/acrylate copolymer, vinyl acetate, crotonic acid copolymer, PVM/MA copolymer, and polyvinylpyrrolidone (PVP), and plasticizers such as acetyl triethyl citrate, benzyl alcohol, and silicones as stiffening agents. Additional ingredients might include silicone, shellac, perfume, lanolin or its derivatives, vegetable gums, alcohol, sorbitol, and water.
- *Hairdressings*, such as pomades, creams, and lotions, give shine and manageability to dry or curly hair. They may be applied to either wet or dry hair and typically consist of lanolin or its derivatives, petrolatum, oil emulsions, fatty acids, waxes, mild alkalis, and water.
- *Masks and packs* are available to serve many purposes and skin conditions, including deep cleansing, pore reduction, tightening, firming, moisturizing, and wrinkle reduction. Clay masks typically contain varying combinations of kaolin (china clay), bentonite, purified siliceous (fuller's) earth or colloidal clay, petrolatum, glycerin, proteins, specially denatured (SD) alcohol, and water. The primary ingredients typically found in peel-off masks are SD alcohol 40, polysorbate-20, and polymers such as polyvinyl alcohol or vinyl acetate.
- *Massage creams* are used to help the hands glide over the skin. They contain formulations of cold cream, lanolin or its derivatives, and sometimes casein (a protein found in cheese).
- *Medicated lotions* are available by prescription for skin problems such as acne, rashes, or other eruptions.
- *Moisturizing creams* are designed to treat dryness. They contain humectants, which create a barrier that allows the natural water and oil of the skin to accumulate in the tissues. This barrier also works to protect the skin from air pollution, dirt, and debris. Moisturizers contain a variety of emollients ranging from simple ingredients such as peanut, coconut, or a variety of other oils to more complex chemical compounds such as cetyl alcohol, cholesterol, dimethicone, or glycerin derivatives.
- *Ointments* are semisolid mixtures of organic substances, such as lard, petrolatum, or wax, and a medicinal agent. No water is present. For the ointment to soften, its melting point should be below body temperature (98.6 degrees Fahrenheit). Ointments are prepared by melting an organic substance and then mixing it with a medicinal agent.
- *Pastes* are soft, moist cosmetics with a thick consistency that are used as a styling aid. They are bound together with the aid of gum, starch, and water. If oils and fats are present, water is absent.

- *Scalp lotions* and ointments usually contain medicinal agents for active correction of a scalp condition such as itching or flakiness. An astringent lotion may be applied to the scalp before shampooing to control oiliness as well as the itching and flakiness of dry scalp conditions. Medicated lotions and ointments for severe scalp conditions must be prescribed by a physician.
- *Soaps* are compounds that are made by mixing plant oils or animal fats with strong alkaline substances. Glycerin is also formed in the process. Potassium hydroxide produces a soft soap, whereas sodium hydroxide forms a hard soap. A mixture of the two alkalis will yield a soap of intermediate consistency. A good soap does not contain excess free alkali and is made from pure oils and fats. Soaps used in the industry may be categorized as deodorant soaps, beauty soaps, medicated soaps, and shaving soaps.
- *Styling aids*, such as gels and mousses, typically consist of polymer and resin formulations that are designed to give the hair body and texture. Many incorporate the same ingredients found in hair sprays but add moisturizers and humectants, such as cetyl alcohol, panthenol, hydrolyzed protein, quats, or a variety of oils to the ingredient list.
- *Suntan lotions* are designed to protect the skin from the harmful ultraviolet rays of the sun. They are rated with a sun protection factor (SPF) that enables sunbathers to calculate the time they can remain in the sun before the skin begins to burn. Suntan lotions are emulsions that might contain para-aminobenzoic acid (PABA), a variety of oils, petrolatum, sorbitan stearate, alcohol, ultraviolet inhibitors, acid derivatives, preservatives, and perfumes.
- *Wrinkle treatment creams* are designed to conceal lines on aging skin either via a crease-filling capacity or through a “plumping up” of the tissues. Among the many possible ingredients in these treatments are hormones and collagen. Some are made of herbs and other natural ingredients while others are entirely synthetic.

REVIEW THE UNITED STATES PHARMACOPEIA

The *United States Pharmacopeia* (USP) is a public health organization that sets standards for food ingredients, healthcare products, and drugs sold or manufactured in the United States and used by the public. Some of the most common chemical ingredients used in the formulation of hair and skin products are as follows:

- *Alcohol* is a colorless liquid obtained from the fermentation of starch, sugar, and other carbohydrates. Isopropyl (rubbing) and ethyl (beverage or grain) alcohol are both volatile (readily evaporated) alcohols. These alcohols function as solvents and can be found in shampoos, conditioners, hair colorants, hair sprays, tonics, and styling aids. Fatty alcohols, such as cetyl and cetearyl alcohol, are nonvolatile oils that are used as conditioners. The alcohols most often used in the barbershop are ethyl alcohol and isopropyl alcohol. Fifty to sixty percent isopropyl alcohol is an effective antiseptic that can be applied to the skin.

- *Alkanolamines* (al-kan-all-AM-eenz) are substances used to neutralize acids or raise the pH of hair products. They are often used in place of ammonia because there is less odor associated with their use. Alkanolamines are used as alkalizing agents in hair lighteners and permanent wave solutions.
- *Alum* is aluminum potassium or ammonium sulfate supplied in the form of crystals or powder; it has a strong astringent action. For this reason, alum is found in some skin tonics and is also used to make styptic powder or liquid to stop the bleeding of small nicks and cuts. Check with your state barber board before using styptic in any form.
- *Ammonia*, a colorless gas composed of hydrogen and nitrogen, has a pungent odor. Ammonia is used to raise the pH in permanent waving, haircoloring, and lightening substances. Ammonium hydroxide and ammonium thioglycolate are examples of ammonia compounds that are used to raise solution pH levels for better penetration into the hair.
- *Formaldehyde* is a colorless gas manufactured by an oxidation process of methyl alcohol. It is used as a disinfectant, fungicide, germicide, and preservative as well as an embalming solution. In the cosmetics industry, small amounts of formaldehyde are used in soaps, cosmetics, and nail hardeners and polishes. Formaldehyde and its derivatives, such as formalin, should be used with caution because the National Cancer Institute studies indicate that it is toxic, can lead to DNA damage, and is known to react with other chemicals to become a carcinogen.
- *Glycerin* is a sweet, colorless, odorless, oily substance formed by the decomposition of oils, fats, or fatty acids. It is used as a solvent and as a skin moisturizer in cuticle oils and facial creams.
- *Hydrogen peroxide* is a compound of hydrogen and oxygen. It is a colorless liquid with a characteristic odor and a slightly acidic taste. Organic matter, such as silk, hair, feathers, and nails, is bleached by hydrogen peroxide because of its oxidizing power. A hydrogen peroxide solution is used as a bleaching agent for the hair in solutions of 20 to 40 volume. A 3 to 5 percent solution of hydrogen peroxide possesses antiseptic qualities.
- *Petrolatum*, commonly known as Vaseline, petroleum jelly, or paraffin jelly, is a yellowish to white, semisolid, greasy mass that is almost insoluble in water. It is used in wax epilators, eyebrow pencils, lipsticks, protective creams, cold creams, and many other cosmetics for its ability to soften and smooth the skin.
- *Quaternary ammonium compounds* (quats) are found in many antiseptics, surfactants, preservatives, sanitizers, and germicides. Quats are synthetic derivatives of ammonium chloride. Although quats can be toxic, they are considered safe in the proportions used in the industry.
- *Silicones* are a special type of oil used in hair conditioners and as water-resistant lubricants for the skin. Silicones are less greasy than plain oils and have the ability to form a “breathable” film that does not cause comedones.
- *Witch hazel* is a solution of alcohol, water, and powder ground from the leaves and twigs of *Hamamelis virginiana*. It works as an astringent and skin freshener. Because of the alcohol content, it should not be applied directly to an open wound or near the eyes.

REVIEW QUESTIONS

- 1 Define organic and inorganic chemistry.
- 2 List the three states of matter.
- 3 Compare elements and atoms, compounds, and mixtures.
- 4 Explain the two ways in which matter can be changed.
- 5 Explain oxidation-reduction (redox) reactions.
- 6 Name the four classifications of chemical compounds.
- 7 Describe the differences between solutions, suspensions, and emulsions.
- 8 Define *pH* and draw a pH scale.
- 9 Explain oxidation and reduction reactions. Give an example of each.
- 10 Why is it important to know the acidity or alkalinity of a shampoo product?
- 11 What effect do acidic solutions have on hair?
- 12 What effect do alkaline solutions have on hair?
- 13 Identify the parts and functions of the shampoo molecule.
- 14 List three types of shampoo formulations.
- 15 List four types of conditioners.

CHAPTER GLOSSARY

acids	p. 184	solutions that have a pH below 7.0
alkalis (AL-kuh-lyz)	p. 184	solutions that have a pH above 7.0
atoms (AT-umz)	p. 180	are the basic building blocks of all matter and are the smallest particle of an element that has the chemical identity of the chemical
balancing shampoos	p. 193	are designed for oily hair and scalps. These shampoos wash away excess oiliness while keeping the hair from drying out
bases	p. 184	see alkalis
chemical change (KEM-uh-kul CHAYNJ)	p. 181	change in the chemical composition of a substance by which new substances are formed
chemical compounds (KEM-uh-kul KAHM-powndz)	p. 184	combinations of two or more atoms of different elements united chemically with fixed chemical composition, definite proportions, and distinct properties
chemical properties (KEM-uh-kul PRAH-ur-teez)	p. 181	characteristics that can only be determined with a chemical reaction
chemistry (KEM-is-tree)	p. 179	the science that deals with the composition, structure, and properties of matter
clarifying shampoos	p. 193	shampoo with an active chelating agent that binds to and helps remove metals from the hair; also provides thorough cleansing and cuts through product buildup
color-enhancing shampoos	p. 193	created by combining surfactant bases with basic dyes
compound molecules (KAHM-pownd MAHL-uh-kyoolz)	p. 180	two or more atoms of different elements united chemically with fixed chemical composition, definite proportions, and distinct properties

conditioners	p. 194	chemical agents used to deposit protein or moisturizers in the hair
conditioning shampoo (KAHN-dish-shun-ing sham-POO)	p. 193	also known as <i>moisturizing shampoos</i> , are mild cream shampoos that contain moisturizing agents (<i>humectants</i>) designed either to “lock in” the moisturizing properties of the product or to draw moisture into the hair
dry or powder shampoos	p. 193	shampoos that cleanse the hair without water
element (EL-uh-ment)	p. 179	is the simplest form of chemical matter and contains only one type of atom
elemental molecules (EL-uh-ment-uhl MAHL-uh-kyoolz)	p. 180	molecules consisting of two or more atoms of the same element in fixed proportions, for example, oxygen O ₂
emulsions	p. 185	mixtures of two or more immiscible substances united with the aid of a binder or emulsifier
endothermic reaction	p. 183	a chemical reaction that requires the absorption of energy or heat from an external source for the reaction to actually occur
exothermic reaction	p. 183	the release of energy in the form of heat through certain chemical reactions
hair tonics	p. 196	grooming aids designed to stimulate the surface circulation of the scalp, remove loose dandruff, or to impart manageability, shine, and control to the hair
inorganic chemistry (in-or-GAN-ik KEM-is-tree)	p. 179	is the study of substances that do not contain carbon but may contain hydrogen
instant conditioners (IN-stant KAHN-dish-uhn-uhz)	p. 194	conditioners that typically remain on the hair from 1 to 5 minutes and are rinsed out; also known as <i>rinse-out conditioners</i>
ion (EYE-ahn)	p. 187	an atom or molecule that carries an electrical charge
ionization (eye-ahn-ih-ZAY-shun)	p. 187	the separating of a substance into ions
leave-in conditioners (LEEV-in KAHN-dish-uhn-uhz)	p. 195	conditioners and thermal protectors that can be left in the hair without rinsing
liquid-dry shampoos	p. 192	liquid shampoos that evaporate quickly and do not require rinsing
matter (MAT-ur)	p. 179	is defined as anything that occupies space (volume) and has mass (weight)
medicated conditioners	p. 195	conditioners formulated to control minor dandruff and scalp conditions
medicated shampoos	p. 193	shampoos containing medicinal agents for controlling dandruff and other scalp conditions
molecule	p. 180	two or more atoms joined chemically
neutralizing shampoo	p. 194	helps to rebalance and restore the pH level of the hair by neutralizing any alkali or unwanted residues that remain in the hair after chemically relaxing the hair
organic chemistry (or-GAN-ik KEM-is-tree)	p. 179	the study of substances that contain carbon
oxidation (ahk-sih-DAY-shun)	p. 182	either the addition of oxygen or the loss of hydrogen; a chemical reaction that combines an element or compound with oxygen to produce an oxide

oxides	p. 184	compounds of any element combined with oxygen
pH (potential hydrogen) (P-H)	p. 188	relative degree of acidity or alkalinity of a substance
pH-balanced shampoos	p. 193	shampoos that are balanced to the pH of hair and skin (4.5 to 5.5); also known as <i>acid-balanced shampoos</i>
pH scale (P-H SKAYL)	p. 188	a measure of the concentration of hydrogen ions in acidic and alkaline solutions
physical change (FIZ-ih-kuhl CHAYNJ)	p. 181	change in the form of a substance without the formation of a new substance
physical mixture (FIZ-ih-kuhl mix-CHUR)	p. 183	combination of two or more substances united physically
physical properties (FIZ-ih-kuhl PRAH-ur-teez)	p. 181	characteristics of matter that can be determined without a chemical reaction
pure substance (PYOOR SUB-stantz)	p. 183	matter that has a fixed chemical composition, definite proportions, and distinct properties
redox (REE-dux)	p. 182	term used for an oxidation-reduction reaction
reduction (ree-DUK-shun)	p. 182	the subtraction of oxygen from, or the addition of hydrogen to, a substance
salts	p. 184	compounds that are formed by the reaction of acids and bases
scalp conditioners	p. 196	cream-based products and ointments used to soften and improve the health of the scalp
shampoo (sham-POO)	p. 190	removes dirt, oil, perspiration, and skin debris from the hair and scalp
solute	p. 185	any substance that is dissolved into a solvent to form a solution
solution	p. 185	a stable, uniform mixture of two or more mixable substances made by dissolving a solid, liquid, or gaseous substance in another substance
solvent	p. 185	any substance, usually a liquid, that dissolves the solute to form a solution
sulfate-free shampoo	p. 194	formulated with little to no sulfates so it does not strip natural oils from the hair
surfactant (sur-FAK-tant)	p. 186	a substance that acts as a bridge to allow oil and water to mix or emulsify by reducing surface tension
suspensions (sus-PEN-shunz)	p. 185	formulations in which solid particles are distributed throughout a liquid medium
therapeutic medicated shampoos (thayr-uh-PYOOT-ik MED-ih-kayt-ud sham-POOZ)	p. 194	contain special chemicals or drugs that are very effective in reducing excessive dandruff
treatment or repair conditioners	p. 195	deep, penetrating formulations that help restore moisture and protein in the hair

8



BASICS OF ELECTRICITY

CHAPTER OUTLINE

INTRODUCTION

WHY STUDY BASICS OF ELECTRICITY

UNDERSTAND ELECTRICITY

PRACTICE ELECTRICAL EQUIPMENT SAFETY

UNDERSTAND TERMINOLOGY ASSOCIATED WITH ELECTROTHERAPY

EXPLAIN LIGHT ENERGY AND LIGHT THERAPY

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

LO1

Define electricity.

LO2

Define common electrical terms and measurements.

LO3

Describe electrical safety devices.

LO4

Examine the modalities a barber might be able to utilize depending on state licensing regulations.

LO5

Explain the electromagnetic spectrum, visible spectrum of light, and invisible light.

LO6

Identify devices used in light-therapy treatments.

Introduction

In today's barbershop, haircutting and other services would be very uncomfortable without electricity. It is probably safe to say that most barbers would not like to return to the days of the hand clipper or a lack of air-conditioning. In this chapter, you will learn some important safety precautions regarding the use of electricity in the barbershop. You will also be introduced to different electrical currents used in scalp and facial electrotherapy treatments.

Light-therapy treatments are another service that barbers can offer their clients. Different light rays have different effects on the skin. For this reason, barbers need to know when it is appropriate to apply the thermal and chemical properties associated with light therapy to the skin.

Electricity is a valuable tool for the barber, provided it is used carefully and intelligently. As you study this chapter, think about all the ways electricity and electrical devices will be used in the barbershop.

why study

BASICS OF ELECTRICITY

Barbers should study and have a thorough understanding of the basics of electricity because:

- Barbers use and rely on a variety of electrical appliances and need to know how to use electricity and electrical devices safely.
- Barbers need to understand that special training is required to safely and correctly repair electrical tools and appliances such as clippers, trimmers, and hot towel cabinets. Improperly repaired tools and appliances can result in blown circuit breakers, electric shocks, fires, and personal injury.
- Barbers need to understand the importance of periodic inspections to minimize the risk of fires caused by faulty wiring or overloaded outlets.

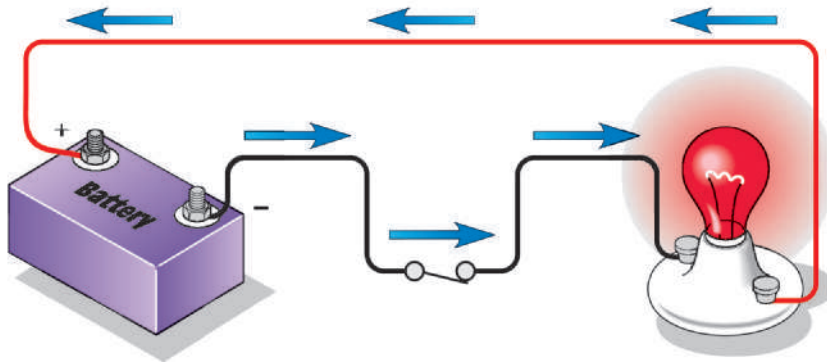
Understand Electricity

After reading this section, you will be able to:

- LO 1 Define electricity.
- LO 2 Define common electrical terms and measurements.

Electricity is not matter because it does not occupy space or have mass; in fact, electricity can move through or across matter and space. Electricity is a form of energy that produces physical, magnetic, chemical, or thermal effects when in motion. This energy is created by the flow of electrons between atoms.

figure 8-1
A complete electric circuit.



- An **electric current** (ee-LEK-trik KUR-ent) is the flow of electricity along a conductor. All substances can be classified as *conductors* or *insulators*, depending on the ease with which an electric current can be transmitted through them.
- A **conductor** (kahn-DUK-tur) is any substance, material, or medium that conducts electricity. Most metals, carbon, the human body, and watery solutions of acids and salts are good conductors of electricity.
- An **insulator** (IN-suh-layt-ur), or *nonconductor* (nahn-kun-DUK-tur), is a substance that does not easily transmit electricity. Rubber, silk, dry wood, glass, and cement are good insulators.
- An *electric wire* is composed of fine, twisted metal threads (the conductor), covered with a silk, plastic, or rubber coating (the insulator).
- A **complete electric circuit** (kum-PLEET EE-lec-trick SUR-kit) is the path of an electric current from the generating source through conductors and back to its original source. **Figure 8-1** illustrates the path of a complete direct-current circuit.



DID YOU KNOW?

Copper is a highly effective conductor used in electric wiring and electric motors.

TYPES OF ELECTRIC CURRENT

There are two kinds of electric current:

- **Direct current (DC)** (dy-REKT KUR-ent) is a constant, even-flowing current that travels in one direction only and produces a chemical reaction. Battery-operated instruments such as flashlights, cell phones, and cordless tools use DC, as does a car battery.
- **Alternating current (AC)** (AWL-tur-nayt-ing KUR-ent) is a rapid and interrupted current, flowing first in one direction and then in the opposite direction, that produces a mechanical action. Electric clippers, hair dryers, and other tools or appliances that plug directly into a wall outlet use AC.

A **converter** (kun-VUR-tur) is an apparatus that changes DC to AC. A **rectifier** (REK-tih-fy-ur) is an apparatus found within a power supply or adapter that converts AC to DC. Rechargeable cordless clippers and battery chargers use a rectifier to convert the AC from an electrical wall

table 8-1

DIRECT CURRENT (DC) AND ALTERNATING CURRENT (AC)

Direct Current	Alternating Current
Constant, even flow	Rapid and interrupted flow
Travels in one direction	Travels in two directions
Produced by chemical means	Produced by mechanical means

outlet to the DC needed to recharge the batteries. A **rheostat** (REE-oh-stat) is an adjustable resistor that is used for controlling the current in a circuit. Many devices, such as light dimmers, use a rheostat.

Table 8-1 outlines the differences between DC and AC.

Electrical Measurements

An electric current flows through a wire in the same way that water flows through a hose; however, without pressure, neither electricity nor water would flow.

- The **volt (V)** (VOLT) is the unit that measures electrical pressure, which pushes the flow of electrons forward through a conductor, similar to the way water pressure pushes water molecules through a hose. Tools and equipment that depend on electricity to function require different voltages; for example, clippers and blowdryers only require the 120 volts in normal wall sockets, while air conditioners and clothes dryers may require 240 volts. A higher voltage indicates more pressure, more force, and more power. The lower the voltage, the weaker the current (Figure 8-2).
- The **amp (A) (or ampere)** (AM-peer) is the standard unit for measuring the strength or rate of an electric current in a conductor. Just as a water hose must be able to expand as the amount of water flowing through it increases, so an electric wire must expand with an increase in the number

of electrons (amps). A cord must be heavy-duty enough to handle the amps put out by the appliance. For example, a hair dryer rated at 10 amps requires a cord that is twice as thick as one rated at 5 amps. If the current, or number of amps, is too strong, the cord can overheat and cause a fire. If the current is not strong enough, the appliance will not operate correctly (Figure 8-3).

- A **milliampere** (mil-ee-AM-peer) is one-thousandth of an ampere. The current for facial and scalp treatments is measured in milliamperes as an ampere current would be much too strong.
- The **ohm (O)** (OHM) is the unit of electrical resistance in an electric current. Unless the force (volts) is stronger than the resistance (ohms), the current will not flow through the wire.

figure 8-2
Volts measure the pressure or force that pushes the electric current forward through a conductor.

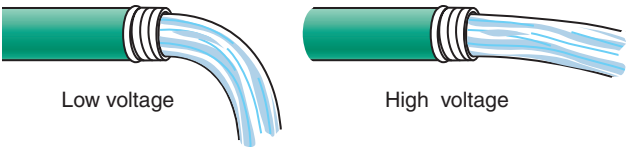
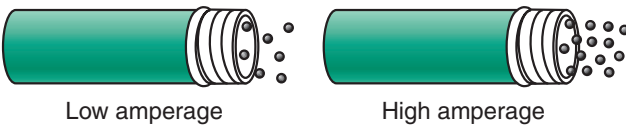


figure 8-3
Amps measure the strength of the electric current.



- The **watt (W)** (WAHT) is the unit of power (amperes multiplied by volts) and indicates how much electric energy is being used in 1 second. A 40-watt bulb, for example, uses 40 watts of energy per second.
- A kilowatt (kW) is 1,000 watts. The electricity in a house is measured in kilowatt-hours (kWh).

Practice Electrical Equipment Safety

After reading this section, you will be able to:

LO 3 Describe electrical safety devices.

When working with electricity, you must always be concerned with your own safety, as well as the safety of your clients. All electrical equipment should be inspected regularly to determine whether it is in safe working order. Careless electrical connections and overloaded circuits can result in electrical shock, a burn, or even a fire.

SAFETY DEVICES

Safety devices help to prevent serious electrical accidents from happening. You should become familiar with these devices and know where they are located in the shop in the event of an emergency or power outage.

- A **fuse** (FYOOZ) is a safety device that prevents the overheating of electrical wires by preventing excessive current from passing through a circuit. The fuse blows or melts when the wire becomes too hot, which happens when the circuit is overloaded with too much current from too many appliances, or if faulty equipment is used. To reestablish the circuit, the appliance must be unplugged or disconnected and a new fuse inserted in the fuse box. Fuse boxes are often found in older buildings that have not been renovated or modernized (Figure 8-4).
- A **circuit breaker** (SUR-kit BRAYK-ar) is a switch that automatically interrupts or shuts off an electric circuit at the first indication of an overload (Figure 8-5). When wires become too hot because of overloading or due to a faulty piece of equipment, the breaker will click off and break the circuit. This action prevents the electric wires and the outlet from overheating and causing a fire. In modern electric circuits, circuit breakers have replaced fuses and can be reset. If an electric appliance malfunctions while in operation, disconnect the appliance from the wall socket immediately and check all connections and insulations before resetting.

Grounding

The principle of *grounding* (GROWN-d-ing) is another important way of promoting electrical safety. All electrical appliances must have at least two electrical connections. One of these connections is neutral and the other is “live” or “hot.” When an appliance is plugged into an outlet, it completes the circuit from the live connection to the neutral connection, producing electricity to operate the appliance. The different sizes of the prongs on modern

figure 8-4
Fuse box.

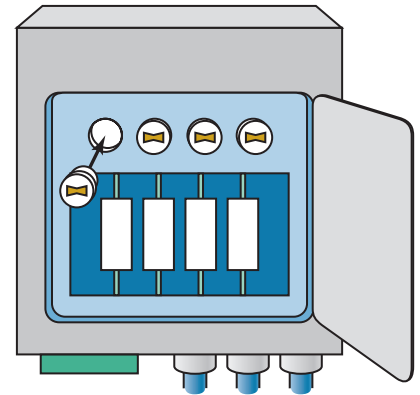


figure 8-5
Circuit breakers.

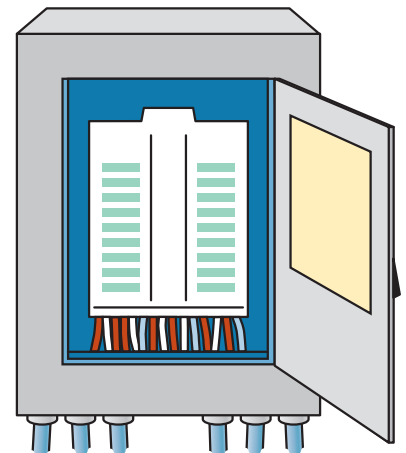


figure 8-6
Two-prong and three-prong plugs.

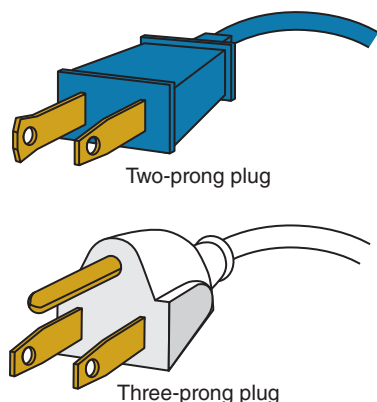


figure 8-7
GFI outlet.

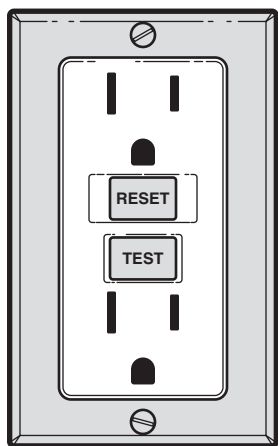


figure 8-8
UL symbol, as it appears
on electrical devices.



electrical plugs guarantee that the plugs can only be inserted one way. This feature provides protection from electrical shock in the event of a short circuit.

Some appliance plugs have a third circular connection called a *grounding pin* that guarantees a safe path to ground for electricity in the event of a short circuit or faulty wiring. This feature provides the most protection for you and your clients, when the power supply of an appliance is encased in metal. Both two-prong and three-prong plugs are grounded at the circuit breaker (**Figure 8-6**).

Ground Fault Interrupter

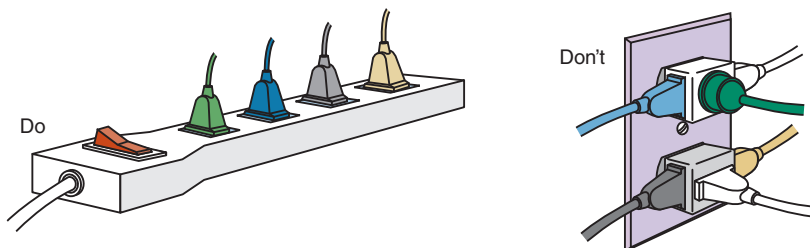
A **ground fault interrupter (GFI)** (GROWND FAWLT in-ter-UP-ter) is an outlet that senses imbalances within an electric circuit (**Figure 8-7**). When it “pops” open, it means that it has sensed a ground fault or current leaking to ground. These devices have surge protectors and must be installed properly to operate as intended; otherwise, although the outlet may work, the protection function will be lost. GFIs are required on all outlets located near or around sinks and water sources in the barbershop.

GUIDELINES FOR SAFE USE OF ELECTRICAL EQUIPMENT

Your safety and the safety of your clients should be of primary concern when you are working with electricity. All electrical equipment should be inspected regularly to determine whether it is in safe working order. Careful attention to electrical safety helps to eliminate accidental shocks, fires, and burns. A review of the following reminders will help ensure the safe use of electricity in the barbershop.

- Check that all electrical appliances you use are UL certified (**Figure 8-8**).
- Study the instructions *before* using any electrical equipment.
- Disconnect appliances when not in use.
- Keep all wires, plugs, and equipment in good repair.
- Inspect all electrical equipment frequently.
- Do not overload outlets and power strips (**Figure 8-9**).
- Avoid getting electrical cords wet.
- When using electrical equipment, protect the client at all times.
- Do not touch any metal while using an electrical appliance.
- Do not handle electrical equipment with wet hands.
- Do not allow the client to touch any metal surfaces while being treated with electrical equipment.
- Do not leave the room while a client is connected to an electrical device.

figure 8-9
Use only one plug per outlet on a power strip or on the wall.



- Do not attempt to clean around an electric outlet while equipment is plugged in.
- Do not touch two metallic objects at the same time if either is connected to an electric current.
- Do not step on, or set objects on, electrical cords.
- Do not allow electrical cords to become twisted or bent; the fine wires inside the cord will break and the insulation will wear away from the wires.
- Disconnect appliances by pulling on the plug, not on the cord.
- Do not attempt to repair electrical appliances! Send the item to the manufacturer or authorized repair dealer to resolve the problem and to maintain your warranty.



CAUTION

Older buildings and homes may have two-prong wall outlets. Some equipment and tools have three-prong plugs. Never tamper with the wiring of the building or home, the wall outlets, or the plugs to make them fit your equipment or tools. Consult the manufacturer or your local hardware store about whether you can use an adapter and, if so, what type of adapter is recommended.

Understand Terminology Associated with Electrotherapy

After reading this section, you will be able to:

- LO 4** Examine the modalities a barber might be able to utilize depending on state licensing regulations.

Electronic facial and scalp treatments are commonly referred to as **electrotherapy** (ee-lek-troh-thair-uh-py). Different types of electric currents are used for facial and scalp treatments. These different types of currents are called **modalities** (MOH-dal-ih-teez) and each modality produces a different effect on the skin. An **electrode** (ee-LEK-trohd) is an applicator used to direct the electric current from the machine to the client's skin. It is usually made of carbon, glass, or metal. Each modality requires two electrodes—one positive and one negative—to conduct the flow of electricity through the body. The only exception to this rule is the Tesla high-frequency current, which is discussed later in this chapter.

DEFINE POLARITY

Polarity (poh-LAYR-ut-ee) indicates the negative or positive pole of an electric current. The electrodes on many electrotherapy devices have one negatively charged pole and one positively charged pole. The positive pole is called the **anode** (AN-ohd), is usually red, and may be marked with a “P” or a plus (+) sign. The negative electrode is called the **cathode** (KATH-ohd). It is usually black and marked with an “N” or a minus (–) sign (**Figure 8-10**). If the electrodes are not marked, contact the manufacturer for more information.

EXPLAIN MODALITIES

Modalities refer to the different types of currents used in electrotherapy treatments. While there are many modalities used in the medical field, this chapter discusses only those modalities that a licensed barber might need. Depending on the rules and regulations in your state, these modalities *may* be galvanic current, microcurrent, and Tesla high-frequency current.

figure 8-10
Anode and cathode.

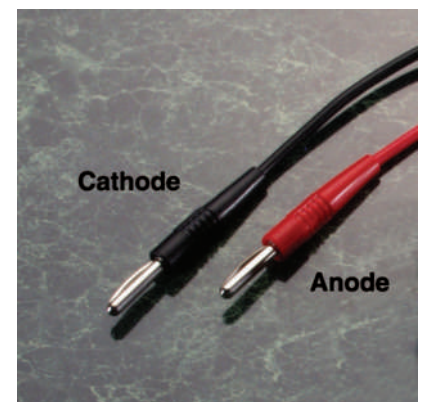


table 8-2

EFFECTS OF GALVANIC CURRENT

Positive Pole (Anode)	Negative Pole (Cathode)
Cataphoresis	Anaphoresis
• Produces acidic reactions • Closes the pores • Soothes nerves • Decreases blood supply • Contracts blood vessels • Hardens and firms tissues	• Produces alkaline reactions • Opens the pores • Stimulates and irritates the nerves • Increases blood supply • Expands blood vessels • Softens tissues



CAUTION

Polarity tests can be dangerous and should not be performed without an instructor’s supervision. Do not allow the tips of the conduction cords to touch, which can cause a short circuit.



DID YOU KNOW?

Galvanic current is named after a doctor named Luigi Galvani, who was born in Italy and lived there until his death in 1798. His studies about electric charges and how they affect the muscles of animals helped others to develop the galvanic current machines used today.



CAUTION

Do not use negative galvanic current on skin with broken capillaries or pustular acne conditions or on clients with high blood pressure or metal implants.

figure 8-11

A microcurrent treatment.



Galvanic Current

Galvanic current (gal-VAN-ik KUR-ent) is a DC, using a negative and positive pole, that is reduced to a safe, low-voltage level. Galvanic current produces chemical changes when passed through body tissues and fluids, and is used to create chemical and ionic reactions in the skin. These reactions depend on the negative or positive polarity of the active electrode that is used on the area being treated. **Table 8-2** shows that the effects produced by the positive pole are the exact opposite of those produced by the negative pole.

- **Desincrustation** (dis-in-krus-TAY-shun) is used to facilitate deep pore cleansing. During this process galvanic current is used to create a chemical reaction that acts to emulsify the sebum and waste in the pores (see Chapter 12).
- **Iontophoresis** (eye-ahn-toh-foh-REE-sus) (meaning “the introduction of ions”) is the process of introducing water-soluble products into the skin. Both the positive and negative poles of the galvanic machine are used in the process. Ionic penetration takes place in two ways:
 - **Cataphoresis** (kat-uh-fuh-REE-sus) infuses acidic (positive) substances into the deeper tissues using the galvanic current from the positive pole toward the negative pole
 - **Anaphoresis** (an-uh-for-EES-sus) infuses alkaline (negative) products into the tissues from the negative toward the positive pole.

Microcurrent

Microcurrent is an extremely low level of electricity that mirrors the body’s natural electrical impulses. Depending on the machine, one or two probes will be used, although some newer devices have both the negative and positive polarities in one probe. Through the probe(s), microcurrent is directed only to the specific area being treated and does not travel throughout the entire body. Microcurrent can be used for iontophoresis, firming, toning, and soothing the skin, and can help to heal inflamed tissue (**Figure 8-11**). Microcurrents can produce the following effects:

- Improve blood and lymph circulation
- Produce acidic and alkaline reactions

- Open and close hair follicles and pores
- Increase muscle tone
- Restore elasticity
- Reduce redness and inflammation
- Minimize healing time for acne lesions
- Improve the natural protective barrier of the skin
- Increase metabolism
- Produce a firmer, more hydrated appearance to aging skin

Tesla High-Frequency Current

Tesla high-frequency current (TES-luh HY-FREE-quens-ee KUR-ent) is characterized by a high rate of oscillation that produces heat. It is commonly called the *violet ray* and is used for both scalp and facial treatments. Tesla current does not cause muscular contractions; its effects are either stimulating or soothing, depending on the method of application.

The electrodes are made of glass or metal and only one electrode is used to perform a service. The shapes of the electrodes vary depending on the service: the facial electrode is flat and the scalp electrode is rake shaped. As the current passes through the glass electrode, tiny violet sparks are emitted (**Figure 8-12**). All treatments given with high-frequency current should be started with mild current and gradually increased to the required strength. Approximately 5 minutes should be allowed for a general facial or scalp treatment, depending upon the condition being treated. As with all other tools and implements, follow the manufacturer's directions. Tesla high-frequency currents may yield the following benefits:

- Stimulate blood circulation
- Improve glandular activity
- Increase metabolism
- Increase absorption of nutrients and elimination of wastes
- Improve germicidal action
- Relieve skin congestion

figure 8-12

Applying Tesla high-frequency current with facial electrode.



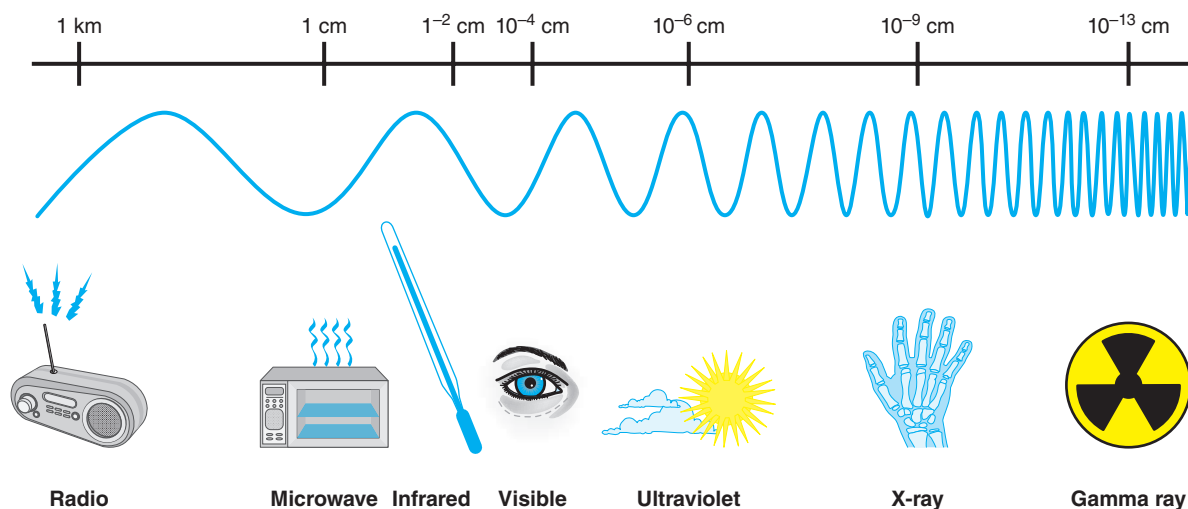
Explain Light Energy and Light Therapy

After reading this section, you will be able to:

- LO 5** Explain the electromagnetic spectrum, visible spectrum of light, and invisible light.
- LO 6** Identify devices used in light-therapy treatments.

Electromagnetic spectrum (ee-lek-troh-MAG-ne-tik SPEK-trum), also known as *electromagnetic spectrum of radiation*, refers to all of the forms of energy (or radiation) that exist. These forms of energy are as follows: radio waves

figure 8-13
The electromagnetic spectrum.



(used by radios and televisions), microwaves (used in microwave ovens), light waves (infrared, visible, and ultraviolet light used for light-therapy services), X-rays (used by physicians and dentists), and gamma rays (used for nuclear power plants) (Figure 8-13).

Energy moves through space on waves. Each type of energy has its own **wavelength** (WAYV-length), the distance between successive peaks of electromagnetic waves. A waveform is the measurement of the distance between two wavelengths. Some wavelengths are long and some are short. Long wavelengths have low frequency, meaning the waves pass a point less frequently within a given length of time. Short wavelengths have a higher frequency because the waves pass more frequently within a given length of time. These differences are compared in Figure 8-14 and Table 8-3.

When light passes through a glass prism, it produces the seven colors of the rainbow, arrayed in the following manner: red, orange, yellow, green, blue, indigo, and violet. Within the visible spectrum of light, violet has the shortest wavelength and red has the longest. In the field of barbering, however, we are more concerned with the invisible rays found at the two ends of the visible spectrum: infrared rays, which produce heat, and ultraviolet rays, which produce chemical and germicidal reactions (Figure 8-15).

figure 8-14
Waveform patterns of long and short wavelengths.

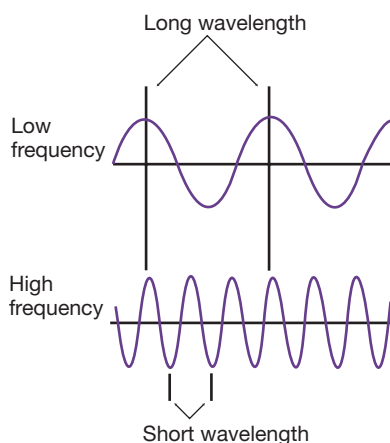
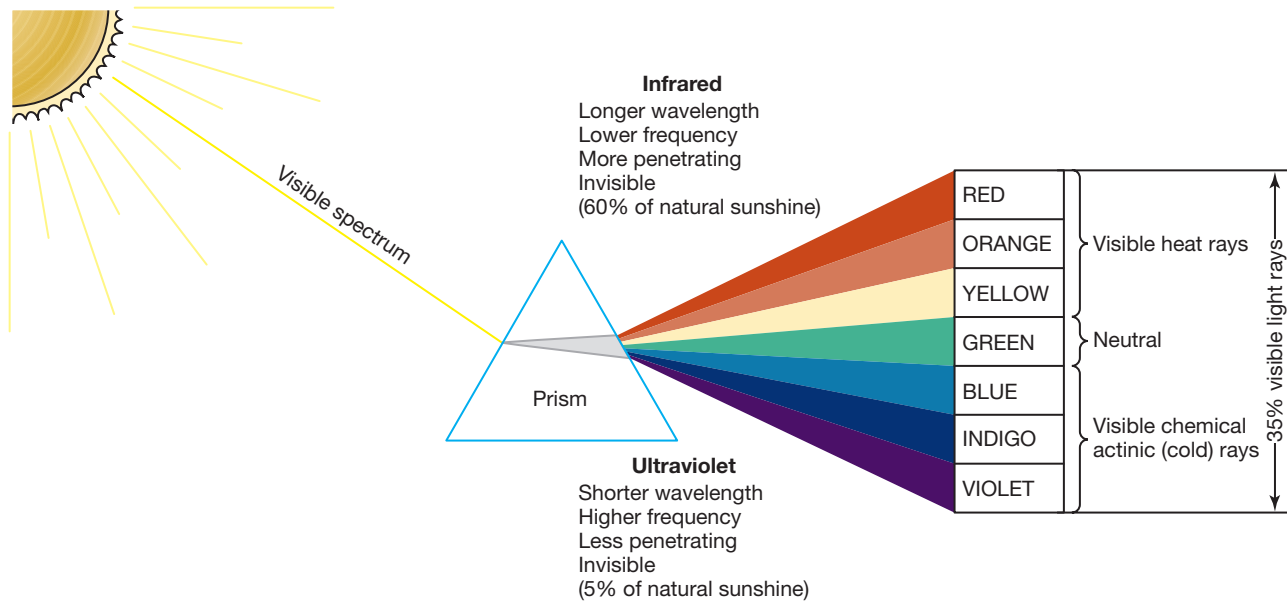


table 8-3

LONG WAVELENGTHS COMPARED WITH SHORT WAVELENGTHS

Long Wavelengths	Short Wavelengths
Low frequency	High frequency
Deeper penetration	Less penetration
Less energy	More energy

figure 8-15
The visible spectrum of light.



THE VISIBLE SPECTRUM OF LIGHT

The **visible spectrum** (VIZ-uh-buhl SPEK-trum) of light is part of the electromagnetic spectrum. Visible light makes up only 35 percent of natural sunlight. Within the visible spectrum of light, violet is at one end of the spectrum and red is at the other end.

INVISIBLE LIGHT

Invisible light is the light at either end of the visible spectrum of light. Ultraviolet rays and infrared rays are forms of electromagnetic energy that are invisible to the human eye because their wavelengths are beyond the visible spectrum of light. Invisible rays make up 65 percent of natural sunlight at the earth's surface.

- **Ultraviolet light** (ul-truh-VY-uh-let LYT), also known as cold light or actinic light, makes up 5 percent of natural sunlight. It has a short wavelength and is less penetrating than visible light. Ultraviolet (UV) rays can kill microorganisms and cause chemical reactions to happen more quickly than visible light while producing less heat. The three types of UV light are:
 - *UVA* rays are closest to the visible spectrum and are the longest of all the UV rays. These rays are used in tanning booths and penetrate deeply into skin tissue. Overexposure can destroy the elasticity of the skin, causing premature aging, wrinkling, and skin cancer.
 - *UVB* rays are in the middle of the UV range, which produce some effects from both ends of ultraviolet rays. Long exposure to UVB rays will burn the skin and, like UVA rays, can cause skin cancer.
 - *UVC* rays are absorbed by the earth's atmosphere and ozone layer and do not reach the surface. However, UVC light rays are used in ultraviolet germicidal irradiation to inactivate or destroy microorganisms.
- **Infrared light** has a longer wavelength when compared to visible light and penetrates more deeply into the body. It produces more heat than

visible light and makes up 60 percent of natural sunlight. Infrared lamps are often used in the barbershop during hair-conditioning treatments to aid product penetration or to accelerate haircolor processing. They are also used in spas to soothe nerves, warm muscles to relieve pain, increase blood circulation, and increase sweat and oil gland production.

LIGHT THERAPY

Light therapy, also known as *phototherapy*, is the application of specific wavelengths of the light spectrum to the skin for the treatment of specific disorders or conditions. In the beauty industry, light-therapy treatments may be accomplished through the use of low-level light therapy or low-level laser therapy, laser (light amplification by stimulated emission of radiation), light-emitting diodes, and therapeutic lamps. There are many light-therapy devices and each one is designed to create specific effects. This is important to know because light changes into electrochemical energy, which creates biochemical reactions in the cells; therefore, different light rays will produce different effects, such as heat, chemical reactions, or germicidal effects in the human body. The most important points to remember are that the equipment to be used is based on the skin type and the condition to be treated and that you will need thorough training in the use of the device.

Depending on the licensing regulations in your state, light therapy may be used to treat a range of skin conditions, from mild eruptions to wrinkles; to treat androgenic alopecia; or to remove unwanted hair. Review the following therapies that you may want to make available to your clients.

LASERS

Lasers (LAY-zurz) are a type of medical device that uses electromagnetic radiation and a medium (solid, liquid, gas, or semiconductor) for hair removal and various skin treatments. Lasers work by *selective photothermolysis* (FOTO-ther-moll-ih-sis), a process that turns the light from a device into heat. Lasers are designed to deliver an intense light beam to a specific depth and to a specific target area without damaging the surrounding tissues. When used in a hair removal treatment the light is converted into heat that is absorbed by the melanin in the follicles, which leads to follicle damage and inhibition of the hair growth. Because most lasers are classified as Level II medical devices or above, an esthetician license and supervision by a qualified physician may be required to operate the laser.

LIGHT-EMITTING DIODE

A **light-emitting diode (LED)** (LYT-EE-mit-ing DYE-ode) is a medical device used to reduce acne, increase skin circulation, and improve collagen content in the skin. The LED works by releasing different colors of light onto the skin to stimulate specific responses at precise depths of the skin tissue. When the colored light reaches a specific depth in the tissue, it triggers a reaction such as stimulating circulation or reducing bacteria. LED treatments should not be performed on anyone who has light sensitivities, has phototoxic reactions, is taking antibiotics, has cancer or epilepsy, is pregnant, or is under a physician's care. Always be certain that your license and state regulations permit the performance of LED treatments.

Depending on the type of equipment used, LEDs use blue, red, yellow, or green light to achieve the following effects:

- Blue light reduces acne and bacteria.
- Red light increases circulation, improves collagen and elastin production in the skin, and stimulates wound healing.
- Yellow light reduces swelling and inflammation, improves lymphatic flow, and detoxifies and increases circulation.
- Green light reduces hyperpigmentation and redness, calms, and soothes.

THERAPEUTIC LAMPS

In the barbershop *therapeutic lamps* may be a more practical solution for delivering light therapies associated with scalp and skin treatments. The lamp usually consists of a dome-shaped reflector mounted on a pedestal with a flexible neck, although several models are available and the bulbs come in various colors for different purposes (Figure 8-16). When using therapeutic lamps, the client's eyes must be protected and should be covered with moistened cotton pads or goggles. A general guide for using therapeutic bulbs follows:

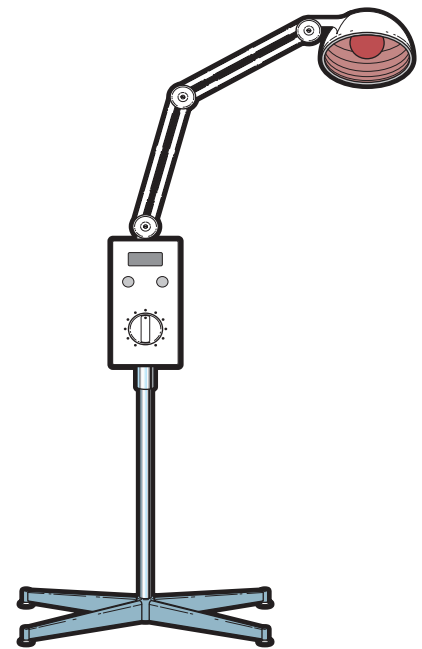
- White light relieves pain, relaxes muscles, and produces chemical and germicidal effects.
- Blue light produces little heat, has a tonic effect on bare skin and a soothing effect on nerves, and is used only on clean, dry skin.
- Red light produces heat rays, creates a stimulating effect, and aids the penetration of creams into the skin.
- Ultraviolet light can be used to treat acne, tinea, seborrhea, and dandruff conditions and is applied at a distance of 30 to 36 inches from the skin. Following are guidelines for using ultraviolet light:
 - The skin area to be treated should be clean and dry.
 - Average exposure can produce skin redness; overdoses will cause blistering.
 - To begin with, a short exposure time of 2 to 3 minutes should be used, with the exposure gradually increasing over a period of days to 7 or 8 minutes.
- Infrared light produces heat but does not produce light; therapeutic lamps using infrared light have only a rosy glow when active and should be applied at an average distance of 30 inches from the skin.
 - Skin product ingredients should be checked for contraindications before use.
 - The light path should be frequently broken, by slowly waving your hand through the path during the service, to maintain client's comfort level.
 - Exposure time should not exceed 5 minutes.

Electrotherapy and light-therapy treatments are special client services that new as well as established barbers should consider offering. When the services discussed in this chapter are performed professionally and marketed effectively, the barbershop will benefit from increased revenue, client retention, and new-client referrals.

**CAUTION**

Never leave the client unattended during light-therapy or laser treatments.

figure 8-16
Therapeutic lamp.



**CAUTION**

The client's eyes should always be protected during light-therapy treatments. Goggles or eye protectors must be provided for clients.

REVIEW QUESTIONS

- 1 Describe two types of electrical current.
- 2 Explain the differences between a fuse, circuit breaker, and ground fault interrupter.
- 3 List and briefly define the three electrical modalities covered in this chapter.
- 4 What does polarity indicate?
- 5 Explain the differences between visible and invisible light rays.
- 6 Explain the differences between ultraviolet and infrared light rays.
- 7 What conditions can be treated with ultraviolet light rays and what is the recommended range of distance between the lamp and the skin?
- 8 Explain the effects of infrared light rays on the skin.
- 9 What is light therapy?
- 10 List four light-therapy treatments used in the beauty industry.

CHAPTER GLOSSARY

alternating current (AC) (AWL-tur-nayt-ing KUR-ent)	p. 207	rapid and interrupted current, flowing first in one direction and then the opposite direction
ampere (AM-peer) or amp (A)	p. 208	standard unit for measuring the strength of an electric current or the rate of flow of charge in a conductor; also called an <i>amp</i>
anaphoresis (an-uh-for-EES-sus)	p. 212	process of forcing substances into tissues using galvanic current from the negative toward the positive pole
anode (AN-ohd)	p. 211	positive electrode
cataphoresis (kat-uh-fuh-REE-sus)	p. 212	process of forcing acidic substances into tissues using galvanic current from the positive toward the negative pole
cathode (KATH-ohd)	p. 211	negative electrode
circuit breaker (SUR-kit BRAYK-ar)	p. 209	switch that automatically interrupts or shuts off an electric circuit at the first sign of overload
complete electric circuit (kum-PLEET EE-lec-trick SUR-kit)	p. 207	the path of an electric current from the generating source through the conductor and back to its original source
conductor (kahn-DUK-tur)	p. 207	any substance, medium, or material that conducts electricity
converter (kun-VUR-tur)	p. 207	an apparatus that changes DC to AC
desincrustation (dis-in-krus-TAY-shun)	p. 212	process used to soften and emulsify oil and blackheads in the hair follicles
direct current (DC) (dy-REKT KUR-ent)	p. 207	constant current that travels in one direction only and produces a chemical reaction
electric current (ee-LEK-trik KUR-ent)	p. 207	the flow of electricity along a conductor
electricity	p. 206	a form of energy that produces physical, magnetic, chemical, or thermal effects when in motion
electrode (ee-LEK-troh-d)	p. 211	an applicator used to direct electric current from a machine to the skin

electrotherapy (ee-lek-troh-thair-uh-py)	p. 211	electronic scalp and facial treatments
electromagnetic spectrum (ee-lek-troh-MAG-ne-tik SPEK-trum)	p. 213	the range of all forms of electromagnetic radiation
fuse (FYOOZ)	p. 209	device that prevents excessive current from passing through a circuit
galvanic current (gal-VAN-ik KUR-ent)	p. 212	constant and a DC, having a positive and negative pole, that produces chemical changes in tissues and body fluids
ground fault interrupter (GFI) (GROWND FAWLT in-ter-UP-ter)	p. 210	a device that senses imbalances in an electric current
infrared light	p. 215	invisible rays with long wavelengths and deep penetration; produce the most heat of any therapeutic light
insulator (IN-suh-layt-ur)	p. 207	substance that does not easily transfer electricity
iontophoresis (eye-ahn-toh-foh-REE-sus)	p. 212	process of introducing water-soluble products into the skin through the use of electric current
laser (LAY-zur)	p. 216	acronym for light amplification by stimulated emission of radiation
light therapy	p. 216	also known as <i>phototherapy</i> , is the application of specific wavelengths of the light spectrum to the skin for the treatment of specific disorders or conditions
light-emitting diode (LED) (LYT-EE-mit-ing DYE-ode)	p. 216	a medical device used to reduce acne, increase skin circulation, and improve collagen content in the skin
modalities (MOH-dal-ih-teez)	p. 211	currents used in electric facial and scalp treatments
ohm (O) (OHM)	p. 208	the unit of electrical resistance to an electric current
polarity (poh-LAYR-ut-ee)	p. 211	negative or positive pole of an electric current
rectifier (REK-tih-fy-ur)	p. 207	apparatus that changes AC to DC
rheostat (REE-oh-stat)	p. 208	an adjustable resistor used for controlling current in a circuit
Tesla high-frequency current (TES-luh HY-FREE-quens-ee KUR-ent)	p. 213	thermal or heat-producing current with a high oscillation rate; also known as the <i>violet ray</i>
ultraviolet light (ul-truh-VY-uh-let LYT)	p. 215	invisible rays, with short wavelengths and minimal skin penetration, that produce chemical effects and kill germs; also called <i>actinic</i> or <i>cold rays</i>
visible spectrum (VIZ-uh-buhl SPEK-trum)	p. 215	electromagnetic spectrum that can be seen by the human eye
volt (V) (VOLT)	p. 208	a unit of electrical pressure that pushes the flow of electrons forward through a conductor
watt (W) (WAHT)	p. 209	the unit of power (amperes multiplied by volts), indicating how much electric energy is being used in 1 second
wavelength (WAYV-length)	p. 214	distance between two successive peaks of electromagnetic waves

9



THE SKIN—STRUCTURE,
DISORDERS, AND
DISEASES

CHAPTER OUTLINE

INTRODUCTION

WHY STUDY THE SKIN—STRUCTURE, DISORDERS, AND DISEASES?

KNOW THE ANATOMY OF THE SKIN

IDENTIFY DISORDERS AND DISEASES OF THE SKIN

DISCUSS DISORDERS OF THE SEBACEOUS AND SUDORIFEROUS GLANDS

RECOGNIZE PIGMENT DISORDERS AND HYPERTROPHIES OF THE SKIN

UNDERSTAND SKIN CANCER

KNOW HOW TO MAINTAIN THE HEALTH OF YOUR SKIN

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

LO1

Describe the structure and divisions of the skin.

LO2

List the functions of the skin.

LO3

Identify and describe common primary and secondary skin lesions.

LO4

Describe common skin inflammations and infections.

LO5

List and describe disorders of the sebaceous and sudoriferous glands.

LO6

List and describe types of skin pigmentations.

LO7

Identify common skin hypertrophies.

LO8

Identify and describe types of skin cancer.

Introduction

As the largest and one of the most important organs of the body, the skin is the body's first line of defense against diseases, protects the organs, and creates a barrier to the environment. **Dermatology** (dur-muh-TAHL-uh-jee) is a branch of medical science that pertains to the study of the skin—its nature, structure, functions, diseases, and treatment. A dermatologist is a physician who specializes in the treatment of the skin, hair, and nails.

why study

THE SKIN—STRUCTURE, DISORDERS, AND DISEASES?

Barbers should study and have a thorough understanding of the skin—structure, disorders, and diseases because:

- Knowing the skin's composition and underlying structures is crucial to performing facials, scalp treatments, and shaving services.
- You will need to recognize and be able to differentiate between normal skin conditions and those that may require medical treatment.
- You will need to recognize abnormal skin conditions that prohibit the performance of certain skin-related services.

Know the Anatomy of the Skin

After reading this section, you will be able to:

- LO 1 Describe the structure and divisions of the skin.
- LO 2 List the functions of the skin.

Healthy skin is slightly moist, soft, and flexible with a smooth, fine-grained texture. The slightly acidic pH of healthy skin provides a protective barrier against organisms that touch or try to enter it. Ideally, the skin should be free of blemishes and other disorders and have the ability to renew itself.

Skin varies in thickness over different parts of the body. It is thinnest on the eyelids and thickest on the palms of the hands and soles of the feet. Continued pressure over any part of the skin can cause it to thicken and become calloused. A callus is part of the body's natural defense system and should not be removed, as it will generally return thicker and harder. The skin of the scalp is constructed similar to the skin elsewhere on the human body, but the scalp has larger and deeper hair follicles to accommodate the longer hair on the head. The appendages of the skin are hair, nails, sweat glands, and oil glands.

The skin is constructed of two clearly defined divisions: the epidermis and the dermis (**Figure 9-1**).

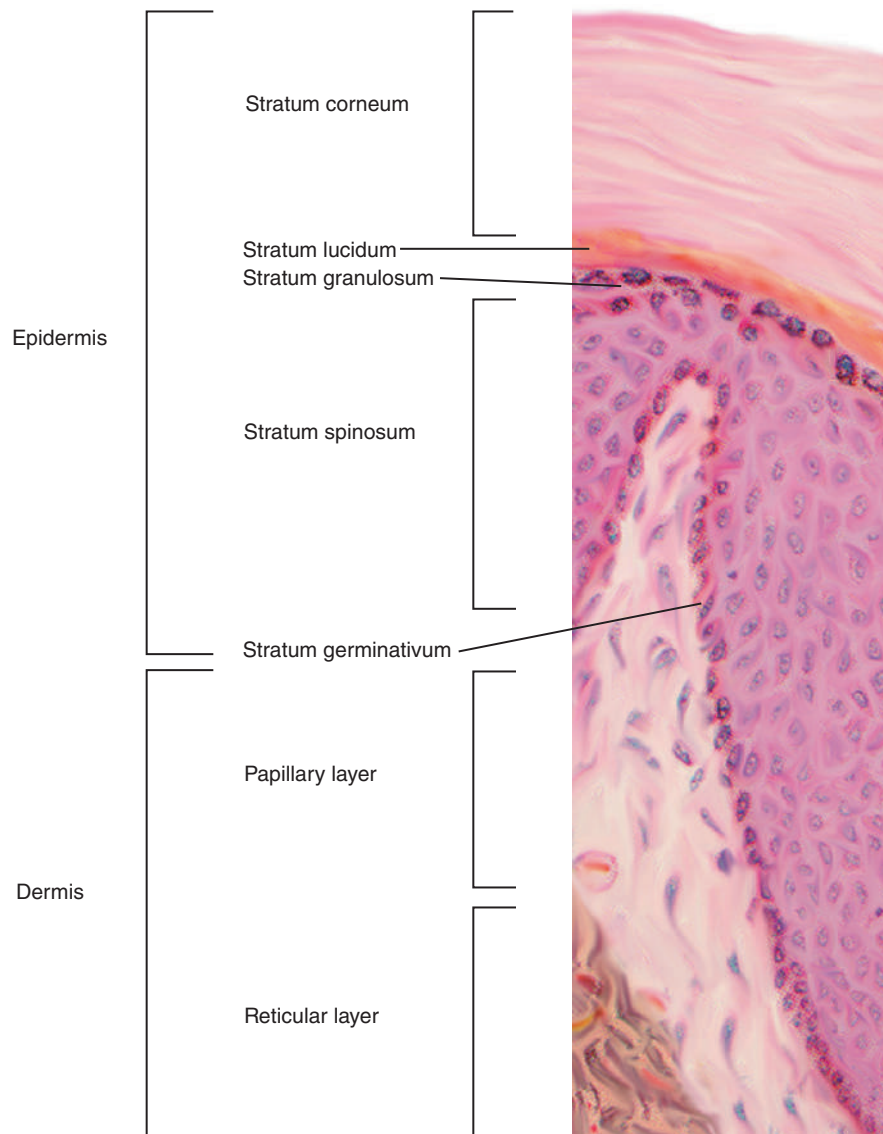


figure 9-1
Layers of the skin.

EPIDERMIS

The **epidermis** (ep-ih-DUR-mus), also known as *cuticle* or *scarf skin*, is the outermost protective layer of the skin and is the thinnest layer of the skin. The epidermis contains no blood vessels, but has many small nerve endings. The layers, or strata, of the epidermis are as follows:

- The **stratum corneum** (STRAT-um KOR-nee-um), or *horny layer*, is the outer layer of the epidermis. This is the layer we see when we look at someone. It consists of tightly packed, scalelike cells that are continually shed and replaced by cells coming to the surface from the underlying layers. These cells are made up of a fibrous protein called *keratin*, which combines with a thin layer of oil (sebum) to help make the stratum corneum a protective, waterproof layer.
- The **stratum lucidum** (STRAT-um LOO-sih-dum), or *clear layer*, lies beneath the stratum corneum and consists of small, transparent cells through which light can pass.

- The **stratum granulosum** (STRAT-um gran-yoo-LOH-sum), or *granular layer*, consists of cells that look like distinct granules and are filled with keratin. These cells are almost dead and are pushed to the surface to replace cells that are shed from the stratum corneum.
- The **stratum spinosum** (STRAT-um spy-NOH-sum), or *spiny layer*, often classified as part of the germinativum, is a sub-layer that lies above the basal strata and beneath the stratum granulosum. The process that causes skin cells to shed begins in this layer.
- The **stratum germinativum** (STRAT-um jur-min-ah-TIV-um), also known as the *malpighian* or *basal cell layer*, is the deepest layer of the epidermis. This layer is responsible for the growth of the epidermis. It also contains melanocytes, which produce a dark pigment called *melanin*, which protects the sensitive cells below from the destructive effects of excessive exposure to ultraviolet light.



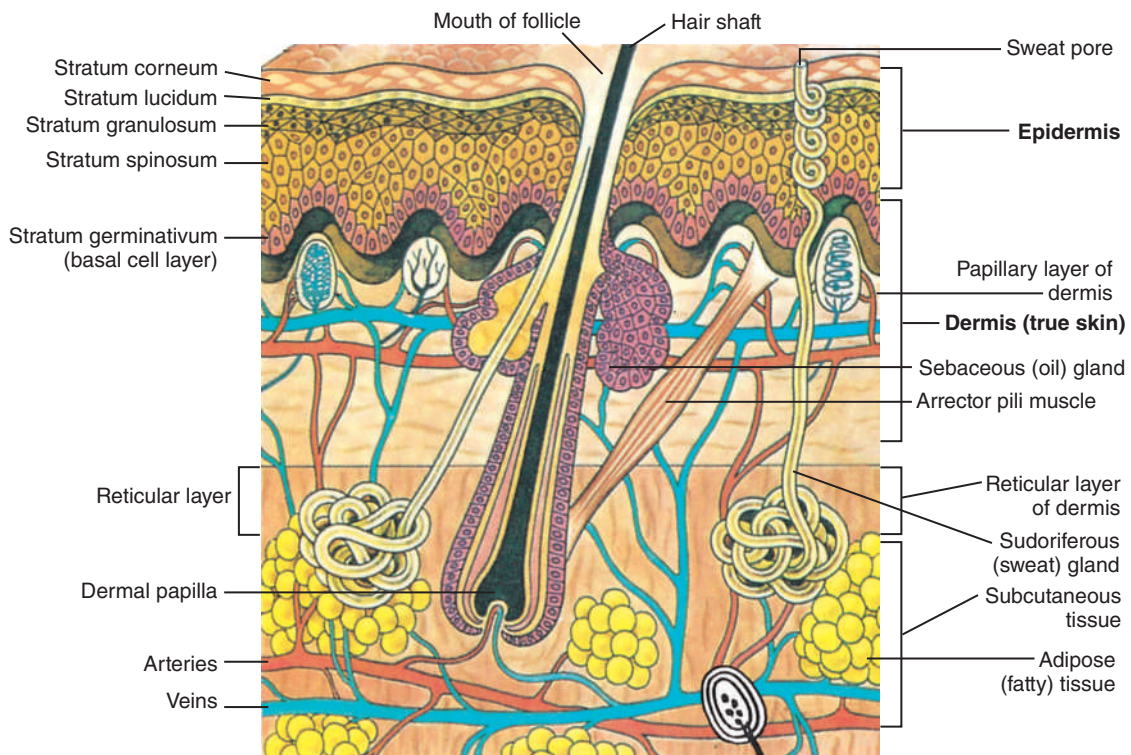
DID YOU KNOW?

Redheads are often called *non-tanners* because they have a variant receptor on their melanocytes that cause them to produce yellow and red hue pigments instead of dark.

DERMIS

The **dermis** (DUR-mis) is the underlying, or inner, layer of the skin. It is also called the **derma** (DURM-uh), *corium*, *cutis*, or *true skin*. The dermis is about 25 times thicker than the epidermis and consists of a highly sensitive vascular layer of connective tissue. Within its structure are numerous blood vessels, nerves, lymph and oil glands, hair follicles, arrector pili muscles, and papillae. The dermis consists of two layers: the papillary or superficial layer, and the reticular or deeper layer (Figure 9-2).

figure 9-2
Structures of the skin.



- The **papillary layer** (PAP-uh-lair-ee LAY-ur) lies directly beneath the stratum germinativum of the epidermis. It contains small, cone-shaped projections of elastic tissue called *papillae* (puh-PIL-eye) that point upward into the epidermis. Some of these papillae contain looped capillaries, which are small blood vessels. Others contain small structures called *tactile corpuscles* with nerve fiber endings that are sensitive to touch and pressure. This layer also contains some melanin (skin pigment). The space at the top of this layer is the epidermal–dermal junction, where the two layers meet.
- The **reticular layer** (ruh-TIK-yuh-lur LAY-ur) is the deeper layer of the dermis, which supplies the skin with oxygen and nutrients. The reticular layer contains the following structures within its network:
 - Fat cells
 - Sweat glands
 - Blood vessels
 - Hair follicles
 - Lymph glands
 - Arrector pili muscles
 - Oil glands

Subcutaneous tissue (sub-kyoo-TAY-nee-us TISH-oo), also known as **adipose tissue** (AD-uh-pohs TISH-oo), is a layer of fatty tissue found below the dermis. Subcutaneous tissue varies in thickness according to age, gender, and general health. It gives smoothness and contour to the body, contains fats for use as energy, and also acts as a protective cushion for the outer skin.

FLUIDS OF THE SKIN

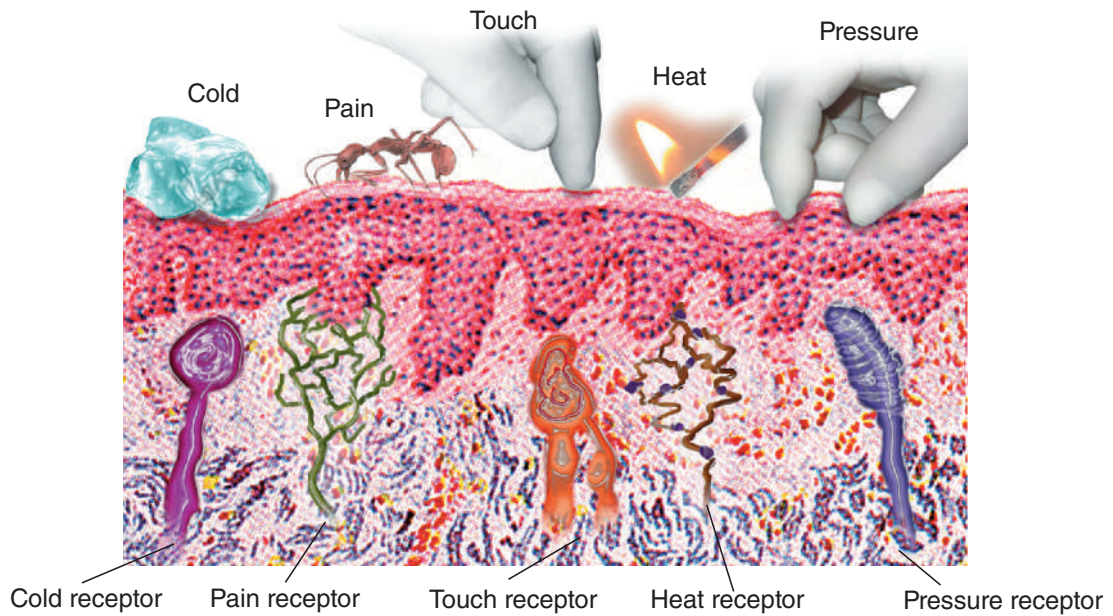
Blood and lymph supply nourishment to the skin in the form of protein, carbohydrates, and fat. From one-half to two-thirds of the body's blood supply is distributed to the skin. As the blood and lymph circulate through the skin, they deliver essential materials for the growth, nourishment, and repair of the skin, hair, and nails. Networks of arteries and lymphatics in the subcutaneous tissue send their smaller branches to hair papillae, hair follicles, and skin glands.

NERVES OF THE SKIN

The skin contains the surface endings of many nerve fibers, which are classified as follows:

- **Motor nerve fibers** (MOH-tur NURV FY-buhrs) are distributed to the arrector pili muscles attached to the hair follicles. These fibers carry impulses from the brain to control muscle movement.
- **Sensory nerve fibers** (SEN-soh-ree NURV FY-buhrs) react to heat, cold, touch, pressure, and pain (**Figure 9-3**). These receptors send messages to the brain.
- **Secretory nerve fibers** (seh-KRUH-toh-ree NURV FY-buhrs) are distributed to the sweat and oil glands of the skin. Secretory nerves regulate the excretion of perspiration from the sweat glands and the flow of sebum from the oil glands to the surface of the skin.

figure 9-3
Sensory nerve endings in the skin.



DID YOU KNOW?

Stimulation such as cold or fear can cause a spontaneous, involuntary contraction of the arrector pili muscles causing the phenomenon known as *goose bumps*, which is the equivalent of *bristling* in fur-covered animals.

Sense of Touch

The papillary layer of the dermis houses the nerve endings that provide the body with the sense of touch. These nerve endings register basic sensations such as touch, pain, heat, cold, pressure, or deep touch. Nerve endings are most abundant in the fingertips. Complex sensations, such as vibrations, seem to depend on the sensitivity of a combination of these nerve endings.

SKIN COLOR

The color of the skin, whether fair or dark, depends on genetics and melanin. Melanin, however, is the primary source of skin color; these grains of pigment are deposited in the stratum germinativum of the epidermis and the papillary layer of the dermis.

Special cells called *melanocytes* produce the pigment granules that are scattered throughout the germinativum and papillary layers. These granules are called *melanosomes* and they produce the complex protein called **melanin** (MEL-uh-nin), a brown-black pigment that gives skin color and serves as the skin's protective screen from the sun's rays. The amount and type of melanin is an inherited trait that varies among races and individuals (**Figure 9-4**). For example, dark skin contains more melanin than light skin.

SKIN ELASTICITY

The skin gets its strength, form, and flexibility from protein fibers within the dermis called *collagen* and *elastin*. **Collagen** (KAHL-uh-jen) fibers make up a large portion of the dermis and help to give support to the many structures found in this layer. When collagen fibers become weakened, wrinkles and sagging of the skin can occur. **Elastin** (ee-LAS-tin) gives the skin its elasticity and flexibility and the ability to regain its shape after stretching. When healthy skin expands, it regains its former shape almost immediately. Conversely, one of the most prominent characteristics of aged skin is its loss of elasticity.

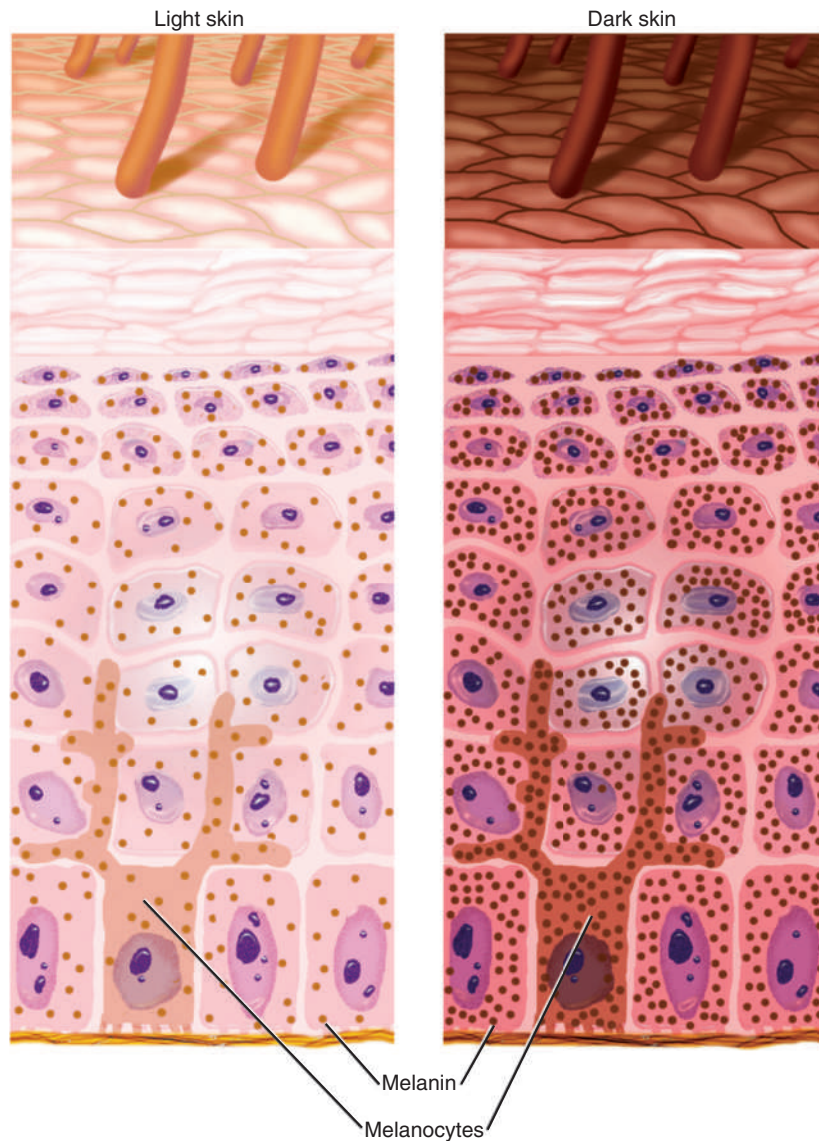


figure 9-4
Melanocytes in the epidermis produce melanin.

THE GLANDS OF THE SKIN

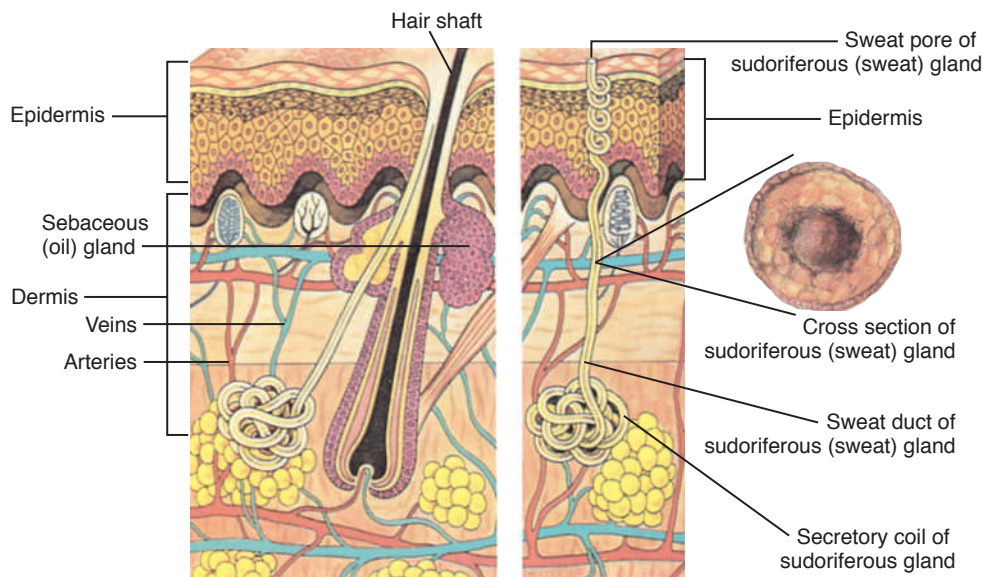
The skin contains two types of duct glands, the *sudoriferous glands*, or *sweat glands*, and the *sebaceous* (sih-BAY-shus) *glands*, or *oil glands*, which extract material from the blood to form new substances (**Figure 9-5**).

Sudoriferous (Sweat) Glands

The **sudoriferous glands** (sood-uh-RIF-uh-rus GLANDZ) consist of a coiled base (called a *fundus*) and a tubelike duct that terminates at the skin surface to form the sweat pore. Practically all parts of the body are supplied with sweat glands, although they are more numerous on the palms, soles, forehead, and armpits.

The nervous system controls the activity of sweat glands, which regulate body temperature and help to eliminate waste products (salt and other chemicals) from the body. Regulation of body temperature occurs when the evaporation of moisture from sweat on the skin cools the body. Heat, exercise, emotion, and certain drugs influence sweat gland activity.

figure 9-5
Sudoriferous glands and the
sebaceous gland.



Sebaceous (Oil) Glands

The sebaceous or oil glands of the skin are connected to the hair follicles. These glands consist of little sacs with ducts that open into the hair follicle, where they secrete **sebum** (SEEB-um), which lubricates the skin and preserves the softness of the hair. With the exception of the palms and soles, these glands are found in all parts of the body, particularly the face and scalp.

Sebum is a semifluid, oily substance produced by the oil glands that flows through the oil ducts leading to the mouths of the hair follicles. When the sebum becomes hardened and the duct becomes blocked, a **blackhead** (BLAK-hed) is formed. The primary function of sebum is to act as a shield that prevents moisture from evaporating from the skin surface.

ABSORPTION LEVEL OF THE SKIN

Although the skin serves as a protective barrier against microorganisms and chemical absorption, some topical products and creams are designed to penetrate this barrier for medicinal purposes. Limited absorption occurs through the skin cells, hair follicles, **sebaceous glands** (seh-BAY-shus GLANDZ), and sudoriferous glands and allows for the entry of specialized drugs or chemicals into the body. For example, antiseptics may be used to reduce the risk of skin infections, or vitamin creams used to support skin repair. The amount or level of absorption that takes place is dependent upon the thickness and location of the skin on the body, the concentration of the product, and the frequency of the application.

DID YOU KNOW?

Many drugs are more effective when the body absorbs them through the skin rather than through the digestive process. Drugs for pain, hormone therapy, and smoking cessation, among others, are safely and effectively delivered through patches that use the absorption properties of the skin to their advantage.

FUNCTIONS OF THE SKIN

The principal functions of the skin are protection, sensation, heat regulation, absorption, excretion, and secretion.

- **Protection.** The skin protects the body from injury and a multitude of pathogens. The outermost layer of the epidermis is covered with a thin layer of sebum, which renders it waterproof. The skin is resistant to variations in temperature, minor injuries, some chemical substances, and many potential pathogens. The skin also provides protection for internal organs, arteries, veins, and bones.

- **Sensation.** The skin responds to heat, cold, touch, pressure, pain, and movement through its sensory nerve endings. Stimulation of a sensory nerve ending sends a message to the brain, which then stimulates a response. When you scratch an itch or pull away from a hot object, you are responding to the stimulation of sensory nerve endings and the message they conveyed to the brain. Some of the sensory nerve endings responsive to touch and pressure lie in close relation to hair follicles.
- **Heat regulation.** Heat regulation is a function of the skin that protects the body from the environment. A healthy body maintains a constant internal temperature of about 98.6 degrees Fahrenheit (37 degrees Celsius). As changes occur in the outside temperature, the blood and sweat glands of the skin make necessary adjustments to facilitate the cooling of the body. Processes such as the evaporation of sweat to cool the body or the closing of pores to retain heat are examples of this adjustment and are regulated by the hypothalamus in the brain.
- **Absorption.** Absorption beyond the top layer of skin is limited, but does occur. Some medications are designed to enter the body through the skin, where they are slowly absorbed and dispersed. Products designed for “skin care” are not intended to be absorbed beyond the top layer of skin.
- **Excretion.** Perspiration is excreted from the skin. Water lost by perspiration carries salt and other chemicals with it. Humans perspire 1 to 2 pints daily; perspiration is made up of water, salt, and chemicals.
- **Secretion.** Sebum is secreted by the sebaceous glands and lubricates the skin and hair, keeping them soft and pliable. Sebum secretion is affected by hormones and emotional stress, which may increase the flow of sebum.

Identify Disorders and Diseases of the Skin

After reading this section, you will be able to:

- LO 3** Identify and describe common primary and secondary skin lesions.

Although barbers are not licensed to perform treatments for medical conditions, providing facials and shaves are traditional services well within the barber's scope of expertise. This includes addressing certain skin conditions, such as oily or dry skin, or alleviating minor acne conditions. Before proceeding with services, barbers need to be able to recognize different types of disorders to determine those that might be aggravated by facials or shaving or those that may need medical attention prior to receiving services.

Some skin and scalp disorders may be treated in cooperation with, or under the supervision of a physician or dermatologist. If asked to apply a medicine to the scalp, you must apply the medicine as directed by the physician and the treatment should not extend beyond the number of applications



HERE'S A TIP

Use the mnemonic SHAPES to remember the functions of the skin: sensation, heat regulation, absorption, protection, excretion, and secretion.



DID YOU KNOW?

Barbers are often the first to recognize changes in their client's skin and scalp! Report any changes in a lesion or growth to the client so he or she may pursue diagnosis and treatment.

or date indicated. To protect both the barber and client, the client should provide the barber with a copy of the prescription directions. If a client has a skin or scalp condition you do not recognize, refer the client to a physician.

LESIONS OF THE SKIN

A **lesion** (LEE-zhun) is a mark on the skin that may indicate an injury or damage that changes the structure of tissues or organs. A lesion can be as simple as a freckle or as dangerous as skin cancer. Lesions can indicate skin disorders or diseases and may be symptomatic of other internal diseases. Being familiar with the principal skin lesions will help you to distinguish between conditions that may be compatible with services and those that need to be referred to a physician prior to the provision of services.

**CAUTION**

To safeguard personal and public health, barbers must not perform services on a client who has (or appears to have) an infectious or contagious disorder. This includes parasitic conditions such as pediculosis and scabies. It is also very important that you do not perform services on inflamed skin, whether it is infectious or not. Certain products and services that are applied on inflamed or irritated skin may intensify and worsen the condition. Barbers should be able to recognize such conditions and tactfully suggest that the client seek appropriate medical treatment before receiving services.

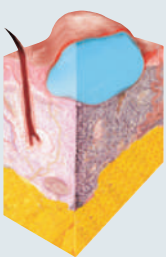
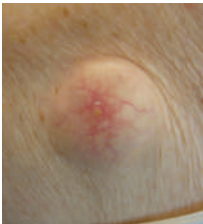
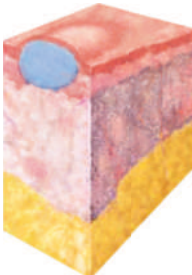
PRIMARY LESIONS OF THE SKIN

Primary skin lesions are lesions that are a different color than the color of the skin or lesions that are raised above the surface of the skin. They are often differentiated by size and layers of skin affected. Such lesions may require medical referral. Refer to **Table 9-1** for a description of primary lesions and examples of each.

SECONDARY LESIONS

Secondary skin lesions are characterized by an accumulation of material on the skin surface, such as a crust or scab, or by depressions in the skin surface, such as an ulcer. Refer to **Table 9-2** for a description of secondary lesions and examples of each.

table 9-1
PRIMARY LESIONS

Primary Lesion	Image	Graphic	Description	Examples
Bulla (BULL-uh) (plural: bullae [BULL-ay])			Large blister containing a watery fluid; similar to a vesicle. Requires medical referral.	Contact dermatitis, large second degree burns, bullous impetigo, perriphigus.
Cyst (SIST) and tubercle (TOO-bur-kul)	 © Courtesy DermNet NZ.		Closed, abnormally developed sac that contains pus, semifluid, or morbid matter, above or below the skin. A cyst can be drained of fluid and a tubercle cannot. Requires medical referral.	Cyst: Severe acne. Tubercle: Lipoma, erythema nodosum.